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Nature and Origin of Zoned Polymetallic (Pb-Zn-Cu-Ag-Au) Veins from the Bingham Canyon Porphyry Cu-Au-Mo Deposit, Utah

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Abstract

Polymetallic veins (Pb-Zn-Cu-Ag-Au) at the world-class Bingham Canyon, Utah, porphyry Cu-Au-Mo deposit have long been recognized, but poorly understood. They are laterally zoned outward from the center of the porphyry deposit transitioning from Fe-Cu to Pb-Zn-Cu-Ag-Au mineralization. Physical and chemical characterization of these polymetallic veins provide insight into the origin, timing, and controls of ore deposition. These sheared, sulfide-rich, NE/SW- trending veins are dominated by pyrite and multiple generations of quartz, with lesser amounts of other sulfide and gangue minerals. Gold (0.27–4.61 ppm) provides the most value to the ore, though the veins contain substantial Cu and Ag as well.

Host rocks include Eocene monzonite and Paleozoic limestone and quartzite—all of which can contain economic ore lodes. Associated alteration is predominantly sericitic and argillic, with mineralization in wall rocks restricted to 1.5 m from the vein margins. Mineral assemblages vary with distance from the center of the main porphyry Cu-Au-Mo deposit and the modal abundances are dependent on the host rock. The appearance of both galena and sphalerite (and tennantite to an extent) occur along a boundary that creates a halo around the center of the associated porphyry deposit. This is accompanied by a shift in metal ratios and an increased concentration of chalcophile trace elements in sulfides from the polymetallic veins as determined by electron microprobe analyses (EMPA) and laser ablation-inductively coupled plasma-mass spectrometry (LA-ICP-MS). Significant hosts of Ag include galena and tennantite, and Cu is hosted primarily in chalcopyrite, tennantite, and sphalerite. The main host of Au could not be determined, but Au could be focused along fractures or hosted in inclusions found in pyrite.

The $\delta^{34}S$ values of vein pyrite have a narrow range (2.3–3.4‰) suggestive of a magmatic source, whereas $\delta^{18}O$ of quartz is more variable (11.5–14.0‰). These values are similar to several other polymetallic vein deposits associated with porphyry Cu deposits. This can be explained by fractionation of magmatic fluids at lower temperatures (350°–250°C) and/or mixing with exchanged ^{18}O-rich meteoric water. Ore grades (Cu, Ag, Au) improve with distance from the center of the porphyry deposit; however, this is accompanied by higher concentrations of deleterious elements (e.g., Pb, As, Bi) for downstream processing.

These polymetallic veins were created sequentially throughout the formation of the deposit. Initial joints in the sedimentary rocks probably formed as a result of emplacement of a barren equigranular monzonite intrusion, with continued dilation and propagation in all host rocks with each subsequent intrusion. The northeast orientation of the joints was controlled by the regional stress field, which is more apparent distal to the center of the Bingham deposit. Vein mineralization appears to postdate all intrusions and the porphyry Cu-Au mineralization; however, it may be related to the late fluids responsible for Mo mineralization in the main porphyry orebody that followed intrusion of the quartz latite porphyry. Quartz-sericite-pyrite mineralization associated with the veins precedes galena-sphalerite-pyrite mineralization. This was followed by late precipitation of chalcopyrite and tennantite and late normal faulting.

Introduction

The Bingham Canyon mine is developed in a porphyry Cu-Au-Mo deposit in the Oquirrh Mountains of northern Utah and centered on an Eocene Igneous Complex. It has been in operation for more than 116 years and has produced greater than 18 million metric tons (Mt) of copper as well as substantial amounts of gold, silver, molybdenum, and sulfuric acid (RioTinto-Kennecott, 2020). Apart from the deposit's massive size, it also contains many of the ore types associated with porphyry Cu systems including a porphyry Cu-Au-Mo deposit, Cu-Au skarns, carbonate-replacement Zn-Pb-Ag-Au, Pb-Zn-Cu-Ag-Au veins, and sediment-hosted Au deposits (Sillitoe, 2010; Fig. 1).

The mine is currently expanding its open-pit operations in an attempt to reach deeper reserves along the southern flank of the pit, where mineralized fissures have been reencountered. The term "fissure" in past studies at Bingham refers to late-stage, large (cm- to m-scale), sulfide-rich veins. In this study, we simply refer to these as zoned polymetallic veins. These veins form ore lodes that vary greatly in size, grade, and geometry even on a single 15-m mine bench, which complicates production predictions. Extensive underground mining exploited these veins on the southern edge of the deposit for Pb, Zn, and Ag as far back as the late 1800s. Today, rather than being mined for Pb, Zn, and Ag, these veins are being mined to supplement the production of Cu, Au, and Ag, of

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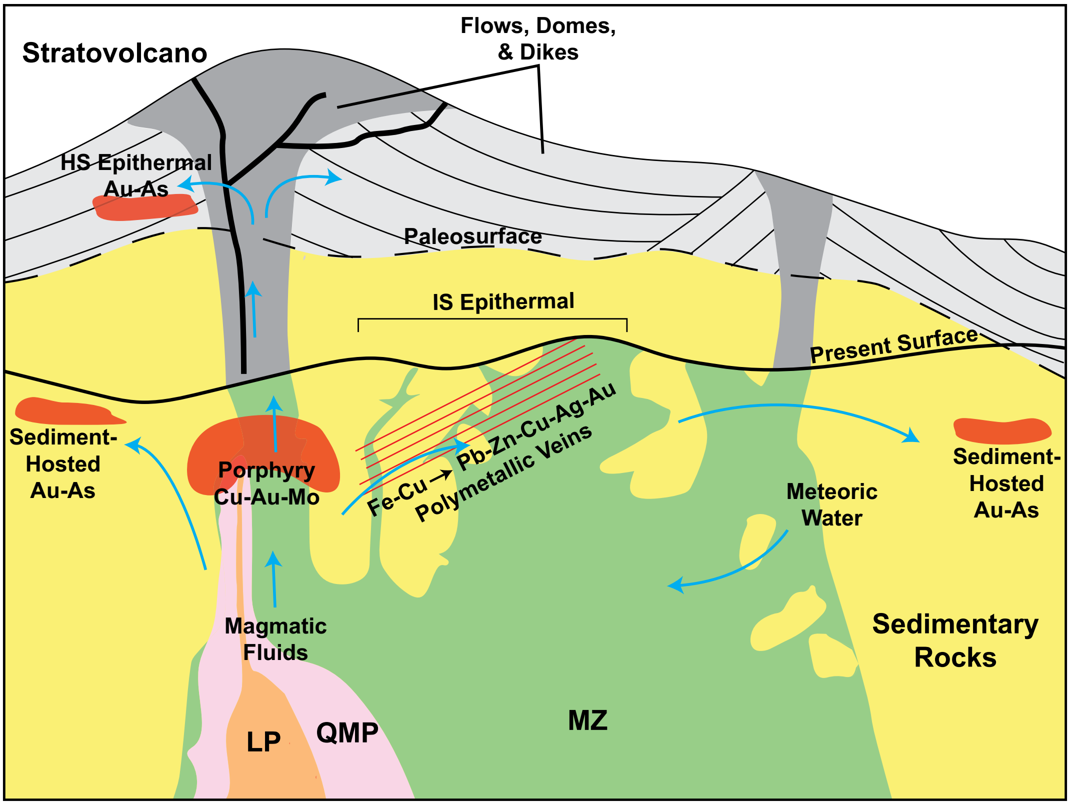

Fig. 1. Simplified cross section of the Bingham porphyry copper system modified from Deino and Keith (1997). Blue arrows indicate fluid pathways, and red shapes and lines represent ore deposits within the system. This study is focused on polymetallic veins at the transition zone from Fe-Cu to Pb-Zn-Cu-Ag-Au mineralization. HS = high sulfidation, IS = intermediate sulfidation, LP = latite porphyry, MZ = monzonite, QMP = quartz monzonite porphyry.

which Au contributes the most overall value. This has sparked a renewed interest in further understanding these veins and their origin.

Porphyry-related polymetallic veins are not unique to Bingham. Other classic examples include the Main Stage veins of the Butte (Meyer et al., 1968), stage 3 veins of the Escondida (Garza et al., 2001), and base metal veins of Morococha (Catchpole et al., 2015). Generalized descriptions, characteristics, and additional examples are outlined by Einaudi et al. (2003) and Fontboté and Bendezú (2009) . These sulfide-rich veins form late in porphyry Cu systems under epithermal conditions (<375°C) and often overprint earlier mineralization with pronounced metal zonation (Fontboté and Bendezú, 2009). At Bingham the mineralization in these veins begins roughly where pervasive porphyry-style mineralization ends (Fig. 1).

There are several previous studies of the Bingham polymetallic veins (Boutwell, 1905; James et al., 1961; Field, 1966; Field and Moore, 1971; Atkinson and Einaudi, 1978), but these are mostly restricted to the base metal mineralization associated with veins beyond active mining operations or in areas where the material has since been removed. This study expands on previous work in order to characterize these veins from a macro- to microscale at the current levels of mining. In addition, we determine possible origins, timing, and controls of mineralization by studying their physical and chemical nature as a function of distance from the center of the deposit.

Regional Geologic Setting

The Bingham deposit is located in northern Utah along the eastern flank of the Oquirrh Mountains. It is aligned with other Eocene ore-related intrusions (Alta stock, Little Cottonwood stock, and stocks associated with the Park City district) along a NE-trending lineament referred to as the Uinta axis (Presnell, 1997; Gruen et al., 2010). The basement rock consists of Paleoproterozoic paragneiss and orthogneiss of the Mojave Province, which were metamorphosed 1.8 to 1.7 Ga (Yonkee and Weil, 2015). Metamorphism was the result of the accretion of the Yavapai Province during the Precambrian assembly of North America (Whitmeyer and Karlstrom, 2007). Arc accretion, granitoid intrusion, and orogenesis associated with the assembly of Rodinia continued until 1.1 Ga followed by rifting during its subsequent breakup in the late Proterozoic (Whitmeyer and Karlstrom, 2007). Accumulation of a wedge of sediment on the subsiding margin in the Pennsylvanian resulted in the formation of the Oquirrh Basin in northwest Utah and deposition of thick successions of marine sediment (Yonkee and Weil, 2015). The basin-filling Oquirrh Group is locally represented by the Butterfield Peaks and the Bingham Mine Formations (Fig. 2). The Mesozoic onset of subduction along the western margin of North America drove the Sevier orogeny and deformed these sedimentary rocks. This created the large folds within the deposit and the Midas thrust beneath the northern end of the deposit (Kloppenburg et al., 2010). The sedimentary formations at Bingham consist

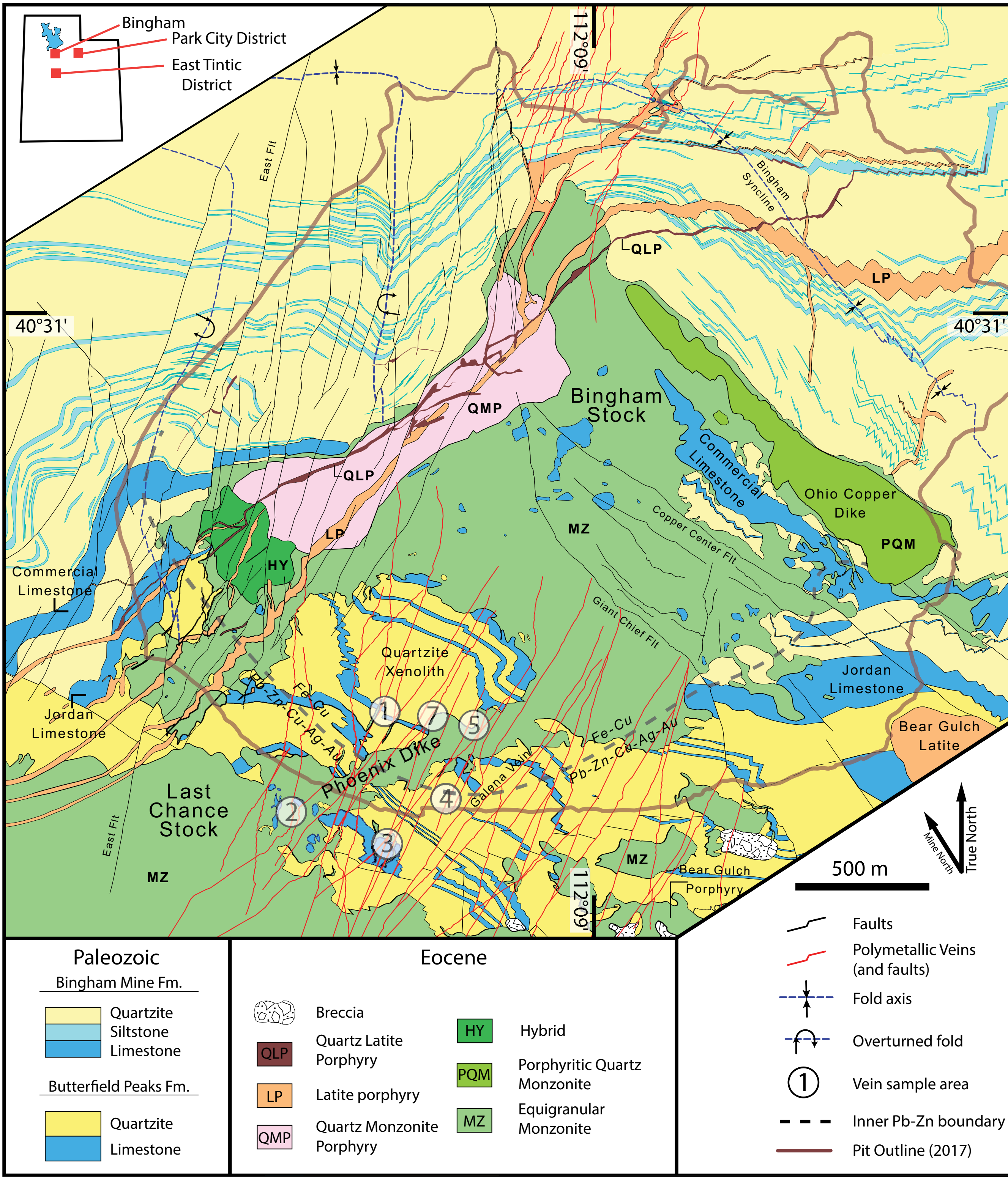

Fig. 2. Geologic map of the Bingham Canyon mine in northern Utah oriented to true north. This study is focused in the southwest portion of the mine where NE-trending, mineralized veins (red) are most abundant. The black dashed line represents where galena and sphalerite mineralization first appear in veins, which also represents the boundary between Fe-Cu and Pb-Zn-Cu-Ag-Au mineralization. Numbers show where samples were taken (sample area 6 was removed as it was inadequate for this study). In the tables, sample names all begin with prefix "FF." The stairstep pattern of the dikes and sedimentary beds in the northeast part of the deposit is the result of mapping dipping units on mine benches. Modified from Porter et al. (2012) with updates provided by mine geologists.

of dominantly thick arkosic orthoquartzite with several distinct siltstone and limestone units (Porter et al., 2012). The Jordan and Commercial limestones represent the base of the Bingham Mine Formation and host high-grade skarns near the contact with the Eocene Bingham intrusive suite (Fig. 2).

The mineralized intrusions can be related to slab roll-back magmatism, which occurred as a result of a progressively steepening dip angle of the subducting Farallon plate during the late Eocene to early Oligocene (Best et al., 2016). The series of intrusions appear to have been emplaced by horizontal dilation on NW- and NE-striking faults (Kloppenburg et al., 2010). These faults have been related to either Eocene extension (Presnell, 1997) or thrusting toward the end of the Sevier orogeny (McKean et al., 2011); they may also be related to the emplacement of the intrusions themselves.

Local Geologic Setting

The timing of the intrusions at the Bingham deposit has long been recognized, but the exact ages have been refined by recent high-precision U-Pb chemical abrasion-isotope dilution-thermal infrared mass spectrometry (CA-ID-TIMS) zircon analyses indicating the ages of the intrusive rocks span from 38.532 to 37.703 Ma (Large, 2018). These ages agree with those derived from $^{40}Ar/^{39}Ar$ chronology by Deino and Keith (1997). The general sequence of intrusions is as follows: equigranular monzonite (38.532 ± 0.043 Ma), quartz monzonite porphyry (38.284 ± 0.024 Ma), latite porphyry (38.129 ± 0.021 Ma), and quartz latite porphyry (37.703 ± 0.057 Ma) (Porter et al., 2012; Large, 2018; Fig. 2). The porphyritic quartz monzonite (Ohio Copper Dike) and hybrid unit have not been adequately dated, but crosscutting relationships suggest they were emplaced soon after the equigranular monzonite and before the quartz monzonite porphyry (Porter et al., 2012).

Cogenetic volcanic rocks exposed just south of the intrusions allow a reconstruction of the paleosurface onto which these rocks were deposited, about 500 m above the premining top of the equigranular monzonite (Deino and Keith, 1997; Fig. 1). It has been suggested that shallow intrusions, such as those exposed at the Bingham Canyon mine, simply act as conduits for magmatic fluid from a deeper and larger magmatic source (Sillitoe, 2010). Although there is no surface exposure of this deeper intrusion, numerical modeling by Steinberger et al. (2013) using aeromagnetic and gravity data requires that a large batholith underlies the base of the current pit at a depth of 2 to 3.5 km with a volume of 1,400 to 3,000 km^3. Magma mixing involving a large volume of intermediate calc-alkaline magma with lesser amounts of mafic alkaline magma contributed to the metal and sulfur supply of the deposit (Hattori and Keith, 2001; Maughan et al., 2002).

Mineralization at the Bingham deposit follows the conventional porphyry Cu-Mo deposit model of a high-grade center with a Cu-barren, Mo-rich core—both of which are associated with potassic alteration. The main Cu mineralizing event occurred between intrusion of the quartz monzonite porphyry and the latite porphyry based on crosscutting vein relationships (Redmond and Einaudi, 2010) lasting for a minimum of 521 ky based on ages determined by Large (2018). Copper-gold and Mo mineralization are zoned spatially and temporally (Landtwing et al., 2010). Seo et al. (2012) suggested a change in fluid chemistry with early Cu-Au-stage fluids being relatively oxidized and neutral-pH, and late Mo-stage fluids being more reduced and acidic. However, there is general agreement that fluid temperature was the primary controlling factor for the overall zonation of the mineralization and alteration (Landtwing et al., 2005; Redmond and Einaudi, 2010). At the Bingham deposit, alteration includes a core of potassic alteration encircled by a large halo of alteration transitional to propylitic alteration with vein- and fracture-controlled sericitic and argillic alteration (Bowman et al., 1987; Porter et al., 2012). Bowman et al. (1987) showed that localized sericitic and argillic alteration are superimposed on both the potassic and propylitic alteration indicating they are late stage.

Our study is focused in the southwest side of the Bingham deposit (Fig. 2), where the dominant ore-related structures are zoned polymetallic veins hosted in units of the equigranular monzonite (Last Chance stock, Bingham stock, and the Phoenix dike), and deformed sedimentary rocks of the Pennsylvanian Butterfield Peaks Formation (primarily quartzite, with lesser amounts of limestone and siltstone). The zonation refers to a shift from inner Fe-Cu mineralization to outer Pb-Zn-Cu-Ag-Au mineralization. A large block of the Butterfield Peaks Formation (which may be a roof pendant) has been completely enclosed by monzonite and is referred to as the "Quartzite Xenolith" (Fig. 2).

The polymetallic veins have a dominant northeast-southwest strike roughly perpendicular to bedding in sedimentary rocks and dip 50° to 90° both northwest and southeast (Fig. 3). The polymetallic veins cut all known igneous units and share a similar northeast orientation as the quartz monzonite

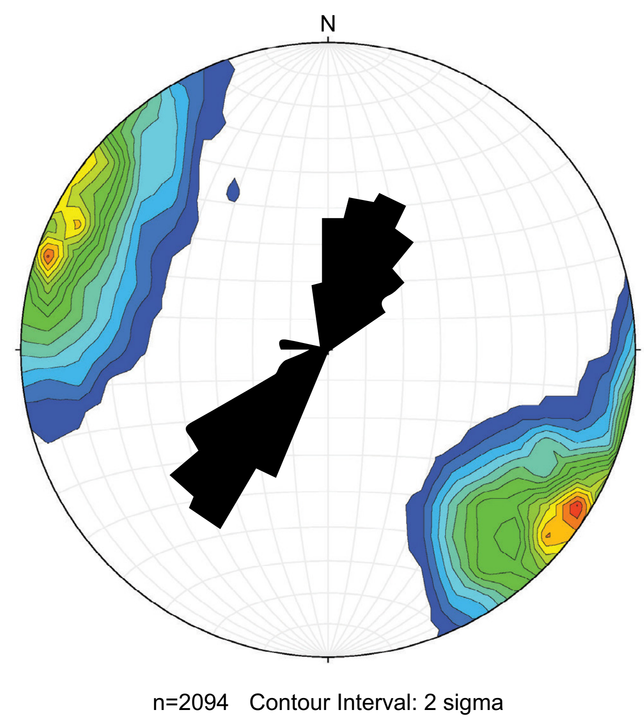

Fig. 3. Orientations of polymetallic veins and associated faults on an equal area stereonet with 2σ Kamb contours from the poles. Orientations are single measurements of mapped veins and associated faults, which are captured on every 15-m mine bench. The average orientation is 77°/035°.

porphyry, latite porphyry, and quartz latite porphyry dikes. Polymetallic veins have also been mapped on the northeast side of the deposit (Fig. 2) with sulfide mineralization typically only within the central porphyry orebody. On the south side of the deposit, the veins begin at the outer extents of the Cu ore shells presented by Porter et al. (2012), but have not been seen to actually cut the porphyry orebody. The polymetallic veins on the north side are also not as long or continuous as those on the south side. This may be related to structural complications with the upright, NW-trending Bingham syncline, which is interpreted to be a Sevier-age buckle fold (Kloppenburg et al., 2010). On the east and west sides of the deposit, bedding plane faults that occur at limestone-quartzite contacts have similar mineralization as the polymetallic veins—the best examples similar to the base of the Jordan and Commercial limestones.

The series of intrusions at Bingham coupled with the regional stress field may have helped form the initial NE-trending structural network that created preferential pathways for hydrothermal fluid flow. On the other hand, it has been proposed that a fluctuating regional stress field between orthogonal orientations repeatedly allowed for the emplacement of the intrusions at Bingham (Gruen et al., 2010; Kloppenburg et al., 2010). It is quite likely that local stress fields formed and changed in response to the sequentially emplaced intrusions, but the consistency of distal dike and polymetallic vein orientations (NE; Figs. 2, 3; Stavast et al., 2006) across the entire deposit suggests they were controlled by a consistent, long-lived regional stress field.

Moreover, the orientation of the NE-trending polymetallic veins is not unique to the Bingham deposit. It is remarkably similar to that of the NE-striking polymetallic veins, pebble dikes, and andesite dikes of the early Oligocene East Tintic mining district 70 km to the south (Hildreth and Hannah, 1996; McKean et al., 2011; Johnson and Christiansen, 2016). Northeast-striking veins and dikes are also associated with several Paleogene plutons along the Uinta axis in the Wasatch Mountains, which includes the polymetallic veins in the Park City district (Presnell, 1997; John, 1989, 1991). They also share similar northeast trends with microfractures, joints, and dikes associated with various late Eocene to early Oligocene plutons in northern Utah and Nevada (Ren et al., 1989; Kowallis et al., 1995). This indicates that although the intrusions were responsible for the formation of these structures—including the Bingham polymetallic veins—a far-field stress controlled their northeasst orientation.

Many of the southern veins are associated with faults with definitive evidence of postmineralization movement based on brecciated and pulverized sulfides within the veins. Some of these faults are continuous along strike from 30 m to more than 2 km and extend as deep as 900 m down from the premine surface (based on current data) and almost certainly continue beyond the open-pit extent. Near the surface, the veins tend to be wider in all rock types and become progressively narrower with depth (Field and Moore, 1971), which also is a feature in late-stage sericitic veins of the Ann-Mason, Nevada, porphyry Cu deposit (Dilles and Einaudi, 1992). Premine topography reconstructed from mapping by mine geologists shows that many of the large polymetallic veins and associated faults formed topographic lows and valleys.

There are several faults within the deposit, but those associated with the polymetallic veins have been grouped together and classified by mine geologists as "Fissure faults." These faults are primarily found in wall rock near the edges of the veins, but occasionally are within the veins themselves. The "East faults" are a group of young faults on the northwest side of the deposit that share the steep dips of the Fissure faults, but more have prominent north-south strikes and lack mineralization (Fig. 2). These appear to be youngest structures in the deposit since they offset some Fissure faults. The southern Fissure faults terminate against the large, lower angle (avg dip of 54°), NW-SE-striking Giant Chief normal fault near the center of the deposit, which suggests the Giant Chief fault likely prevented the vein-related stress from being transmitted. Faults, like the Giant Chief, can have sulfide mineralization at intersections with the polymetallic veins forming "pockets" of mineralization along the strike of the fault.

The dominant alteration type in the southwest wall of the pit is propylitic; it is most obvious in the monzonite where it is marked by secondary actinolite, chlorite, epidote, and calcite. The alteration is pervasive but is most evident along joint surfaces as dark-green coatings. The actinolite replaces pyroxene, hornblende, and locally biotite (Porter et al., 2012). The polymetallic veins cut the propylitic alteration and have structurally controlled sericitic and argillic alteration near the vein margins, as noted by Bowman et al. (1987).

Sampling and Analytical Methods

Six veins were selected for in-depth study in the southwest wall of the Bingham pit (Fig. 2; Table 1). Care was taken to select representative veins in each of the major host rocks. These were supplemented with data from other veins and associated faults collected by mine geologists in order to ensure proper characterization. In addition to sampling the fill material of each vein, wall-rock samples were taken on either side at 0.3-, 1.5-, 3.0-, and 6.0-m distances. Thin sections from each vein were made along with one at each of the intervals mentioned previously. The rocks chosen to make the thin sections were meant to be representative with preference given to samples that contained sulfide veinlets related to main vein mineralization.

Whole-rock analysis was completed by ALS Global in Vancouver, British Columbia, by fire assay and atomic absorption spectroscopy, inductively coupled plasma-atomic emission spectroscopy (ICP-AES), and inductively coupled plasma-mass spectrometry (ICP-MS).

Electron microprobe analyses (EMPA) of pyrite, chalcopyrite, tennantite, galena, and sphalerite were performed on a Cameca SX50 electron microprobe at Brigham Young University. The analytical conditions were 20 kV, 10-40 nA, and count time of 20-60 seconds. The following natural standards were used: FeS_2 (Fe, S), $CuFeS_2$ (Cu, Fe, S), GaAs (As), ZnS (Zn), and PbS (Pb).

These same grains were chosen for additional analysis by laser ablation ICP-MS (LA-ICP-MS) and inspected by a combination of optical microscopy, scanning electron microscopy (SEM), and EMPA to ensure the grains were free of inclusions and cracks. Thin sections were used as opposed to grain mounts in order to preserve textural relationships. Sulfides in polished thin sections were analyzed on a Teledyne Analyte

Table 1. Details of Polymetallic Veins Selected for Study

Vein	Host rock	Rock Unit	Width (cm)	Dip/strike	Elevation (m)	Fault product(s)	Avg distance between veins (m)[1]	Mineral assemblage[2]
FF1	QZ	ILSU quartzite	15	86°/38°	1917	Breccia, powdered qz and sulfides	8.2	py>qz>ser>(ccp)
FF5	MZ	Phoenix dike	15	86°/358°	1902	Clay	7.3	py>qz>cal>ser>ccp>tnt
FF7	MZ	Phoenix dike	150	74°/239°	1902	Clay and powdered sulfides	8.6	py>qz>(ccp)
FF2	MZ	Phoenix dike/ Last Chance stock	10	78°/194°	2115	Clay	8.3	py>ser>ccp>tnt>qz>gn>(ap)
FF3	LS	A-bed	20	81°/199°	2115	Breccia	8.1	py>gn>sp>cal>ccp>qz>tnt>ser
FF4	QZ	Highland quartzite	10	83°/236°	2070	Breccia, powdered qz and sulfides	4.3	py>qz>ab>ser>(ccp)>(gn)>(sp)>(ap)

Notes: All sedimentary units are part of the Butterfield Peaks Formation; abbreviations: ILSU = intermediate upper limestone, LS = limestone, MZ = monzonite, QZ = quartzite; mineral assemblage abbreviations: ab = albite, ap = apatite, cal = calcite, ccp = chalcopyrite, gn = galena, py = pyrite, qz = quartz, ser = sericite, sp = sphalerite, tnt = tennantite

[1] Determined by counting number of veins per 200 m near sample area

[2] Minerals in parentheses were only found as inclusions in other minerals

Excite 193-nm excimer laser ablation system coupled to an Agilent quadrupole ICP-MS (7500 ce series) at the University of Utah. The ICP-MS was tuned while rastering across NIST610 glass prior to each session. Laser ablation conditions varied depending on the minerals being analyzed. Each spot was preablated with two shots to remove contaminants and the spot size was 40 to 50 μm. The frequency varied between 5 to 7 Hz, and fluency ranged 5 to 7 J cm^{-2}. The total analysis time was 90 s for pyrite (20 s background, 70 s ablation) with a 30-s washout time in between analyses. For chalcopyrite, sphalerite, tennantite, and galena the analysis time was 70 s (30 s background, 40 s ablation) with a 30-s washout time in between analyses. Each pyrite grain was analyzed at both the core and the rim. For chalcopyrite, sphalerite, tennantite, and galena, 1 to 2 spots per grain were analyzed, but not always in consistent locations due to the amount of inclusions, fractures, and overall smaller grain size of these minerals. Following the precautions of Cook et al. (2016), tennantite and galena were analyzed last to avoid high background counts of As and Pb in subsequent analyses. The following isotope suite was analyzed: ^{34}S, ^{51}V, ^{52}Cr, ^{55}Mn, ^{57}Fe, ^{59}Co, ^{60}Ni, ^{65}Cu, ^{66}Zn, ^{69}Ga, ^{75}As, ^{78}Se, ^{95}Mo, ^{107}Ag, ^{111}Cd, ^{115}In, ^{118}Sn, ^{121}Sb, ^{125}Te, ^{182}W, ^{193}Ir, ^{197}Au, ^{202}Hg, ^{205}Tl, ^{208}Pb, and ^{209}Bi. The dwell times varied with 0.05 s for Au, 0.005 to 0.02 s for major elements, and 0.02 s for all other elements.

MASS-1, a pressed sulfide powder, served as an external standard and was analyzed three times before and after each set of 10 analyses, or when moving to a different mineral. The concentrations of elements from the most recent USGS certificate of analysis were used for all elements except Au. Gold has been removed from the certificate of analysis after discovering it was heterogeneously distributed in MASS-1 (S. Wilson, pers. commun., 2018), which was observed during line scans. The original concentration of Au from Wilson et al. (2002) was used for MASS-1 during data reduction. Reported values of Au in this study should therefore be taken with caution. The laser ablation data were reduced using Iolite v2.5 (Paton et al., 2011) with internal standard values determined by separate electron microprobe analysis.

Sulfur and oxygen isotope analyses were performed on pyrite and quartz separates from each of the veins. These minerals were selected as they were abundant and ubiquitous in each vein regardless of location. The mineral separates were analyzed by the Queens Institute for Isotope Research in Kingston, Canada. The sulfur isotope composition was measured using a MAT 253 Stable Isotope Ratio Mass Spectrometer coupled to a Costech ECS 4010 Elemental Analyzer with δ^{34}S reported relative to the Vienna Canyon Diablo Troilite (VCDT). Oxygen was extracted following methods of Clayton and Mayeda (1963) and analyzed on a Thermo-Finnigan Delta$^{\text{Plus}}$ XP Isotope Ratio Mass Spectrometer with values reported relative to the Vienna Standard Mean Ocean Water (VSMOW).

Physical Characteristics of Polymetallic Veins

Characteristics of the polymetallic veins were studied on both a macro- and microscopic scale, which ranged from the mine extent down to those only observable by SEM imaging. There are similarities between all of the veins, which is why they were initially grouped together over a century ago (Boutwell, 1905). Nonetheless, variations in structures, textures, and mineral assemblages exist over a wide range of scales. These are most closely related to the host rock and proximity to the center of the Bingham deposit (Tables 1, 2).

Macroscopic physical characteristics of polymetallic veins in all host rocks

The polymetallic veins are 1 cm to greater than 1.5 m wide and filled with sulfide and gangue minerals (Fig. 4). The vein width can vary considerably even from one mine bench to another. Smaller sulfide veinlets (<1 cm) are common in the adjacent wall rock and are both parallel and perpendicular to the main veins (Fig. 4A). Sulfide mineral abundance in these veinlets and wall rock decreases sharply with increasing distance from the mineralized veins. Several prominent, unmineralized joint sets are found in conjunction with these veins with one set always roughly parallel to the vein orientation. Faulting along the polymetallic veins was normal based on the offset of sills, and slightly oblique as evidenced by inclined slicken lines. Offset is typically not more than a few meters, but some of the largest faults such as the Niagara fault have >50 m of offset based on recent modeling by mine geologists.

Table 2. General Characteristics of Polymetallic Veins by Host Rock and Metal Zone

Host rock	Quartzite		Monzonite		Limestone	
Metal zone[1]	Fe-Cu	Pb-Zn-Cu-Ag-Au	Fe-Cu	Pb-Zn-Cu-Ag-Au	Fe-Cu	Pb-Zn-Cu-Ag-Au
Sulfides/sulfosalts	py>ccp	py>ccp>sp>gn	py>ccp>(tnt)	py>ccp>tnt>mrc>gn>sp	py>ccp>gn>sp	py>gn>sp>ccp>tnt
Gangue minerals	qz>ser	qz>ser>cal	qz>ser	ser>qz>cal	qz>cal>ser	cal>dol>qz>ser
Intensity of brecciation	High		Low		Med	
Wall-rock alteration	Weak silic. + weak ser.		Moderate silic. + strong ser. + strong arg.		Strong silic.	
Lateral extent of wall-rock alteration	<0.3 m		<1.5 m		<1 m	
Veinlet mineralization	py>qz>ser		py>qz>ser	py>qz>ser>cal	qz>py>cal	ser>py>gn>sp>cal
Lateral extent of sulfides in veinlets	>1.5 m		<3 m		<3 m	
Examples	Figs. 4a, 5b, 6a, d, e, f		Figs. 4b, d, 5a, c, 6c, g, i, j		Figs. 4c, 5d, 6b, h	

Notes: These characteristics are largely for the extent of our study area and may not be relfective of the entire length of these polymetallic veins; abbreviations: arg. = argillization, cal = calcite, ccp = chalcopyrite, dol = dolomite, gn = galena, mrc = marcasite, py = pyrite, qz = quartz, ser. = sericitization/sericite, silic. = silicification, sp = sphalerite, tnt = tennantite

[1]Contact between metal zones shown in Figure 2 as Pb-Zn boundary

Faults are especially evident where monzonite dikes, which parallel the polymetallic veins, are juxtaposed against sedimentary rocks.

Macroscopic physical characteristics of polymetallic veins variable by host rock

The polymetallic veins vary depending on the host rock types (Table 2). Simple line scans yield an average spacing of 7.5 m between the veins in all rock types, but in the quartzites this spacing is as small as 4.3 m (Table 1). Quartzite appears to be more brittle than monzonite or limestone, which causes more intense brecciation in the veins and proximal wall rocks (Fig. 4A). The result is a higher spatial density and more intense vein "swarms" in the quartzite relative to the monzonite and limestone (Fig. 4D). Additional fault products include pulverized wall rock and sulfides, which can range anywhere from sand- to clay-sized particles (Table 1).

Wall-rock alteration results in a light-colored selvage due to an abundance of quartz, sericite, and sometimes calcite in the adjacent monzonite (Fig. 4B) and limestone (Fig. 4C). Quartzite appears largely unchanged at the outcrop scale (Fig. 4A) except where the veins pass through oxidized quartzite units, which have lighter colored selvages. Vein fill in the quartzite is dominated by pyrite and quartz. The veins in monzonite additionally have chalcopyrite, sericite, and clay, whereas the limestone-hosted veins have pyrite, chalcopyrite, galena, sphalerite, and calcite (Fig. 5). Other mineralogical variations exist at the microscopic scale and are discussed in the following section.

Microscopic physical characteristics of polymetallic veins in all host rocks

Many microscopic-scale characteristics are common in all of studied polymetallic veins regardless of host rock or distance from the center of the deposit. The dominant sulfide is euhedral pyrite, which coexists with a preferred interstitial gangue mineral that varies between veins. The sulfides are generally <1.5 mm across and average 0.5 mm. Sulfides are strongly fractured and healed by gangue minerals, and less commonly by other sulfides, such as chalcopyrite and tennantite. The microfractures parallel the veins, creating microbreccia "channels" through the fill cemented by later generations of primarily quartz and sulfides (Fig. 6A). Mineral inclusions are present in all sulfides and frequently form "trails" that mimic mineral growth zones on the outer edges of their hosts (Fig. 6B). Vugs (possibly fluid inclusions) are also present in pyrite grains, though their distribution within grains is more random. Chalcopyrite and quartz inclusions (~10–20 μm) are hosted by pyrite grains in all vein samples (Fig. 6C). Sulfide mineralization in the host rocks is restricted to small veinlets that decrease in size, sulfide abundance, and quantity with increasing distance from the main vein. Most sulfides disappear completely from these veinlets >3 m from the main veins, at which point nonsulfide gangue minerals dominate.

Three generations of quartz are present, as determined by crosscutting relationships, grain size, and SEM-CL imaging (Fig. 7). The first generation is found near the wall-rock selvage. It has euhedral grains, but SEM-CL imaging reveals resorbed and embayed, bright cores with dull, euhedral rims. The second generation of quartz forms clusters of euhedral, oscillatory zoned crystals intergrown with pyrite. The third generation is very fine grained and restricted to small interstitial veinlets that have healed the fracture networks (Fig. 6E). Networks of smaller, secondary quartz veinlets are also found in the main orebody and are associated with late Cu-Fe sulfide precipitation (Landtwing et al., 2005).

Chalcopyrite and tennantite grains not found as mineral inclusions are always anhedral and intergrown with one another (Fig. 6I). Their textures indicate they were late stage relative to the other sulfides and that tennantite is often a replacement product of chalcopyrite (Fig. 6I, J). They are concentrated along healed fractures and also fill interstitial space. The replacement of chalcopyrite by tennantite also occurs in the late-stage alteration related to skarns in the Commercial and Jordan limestones on the west side of the Bingham deposit (Atkinson and Einaudi, 1978). This replacement relationship is consistent with textures and mineral assemblages associated with an intermediate-sulfidation environment and

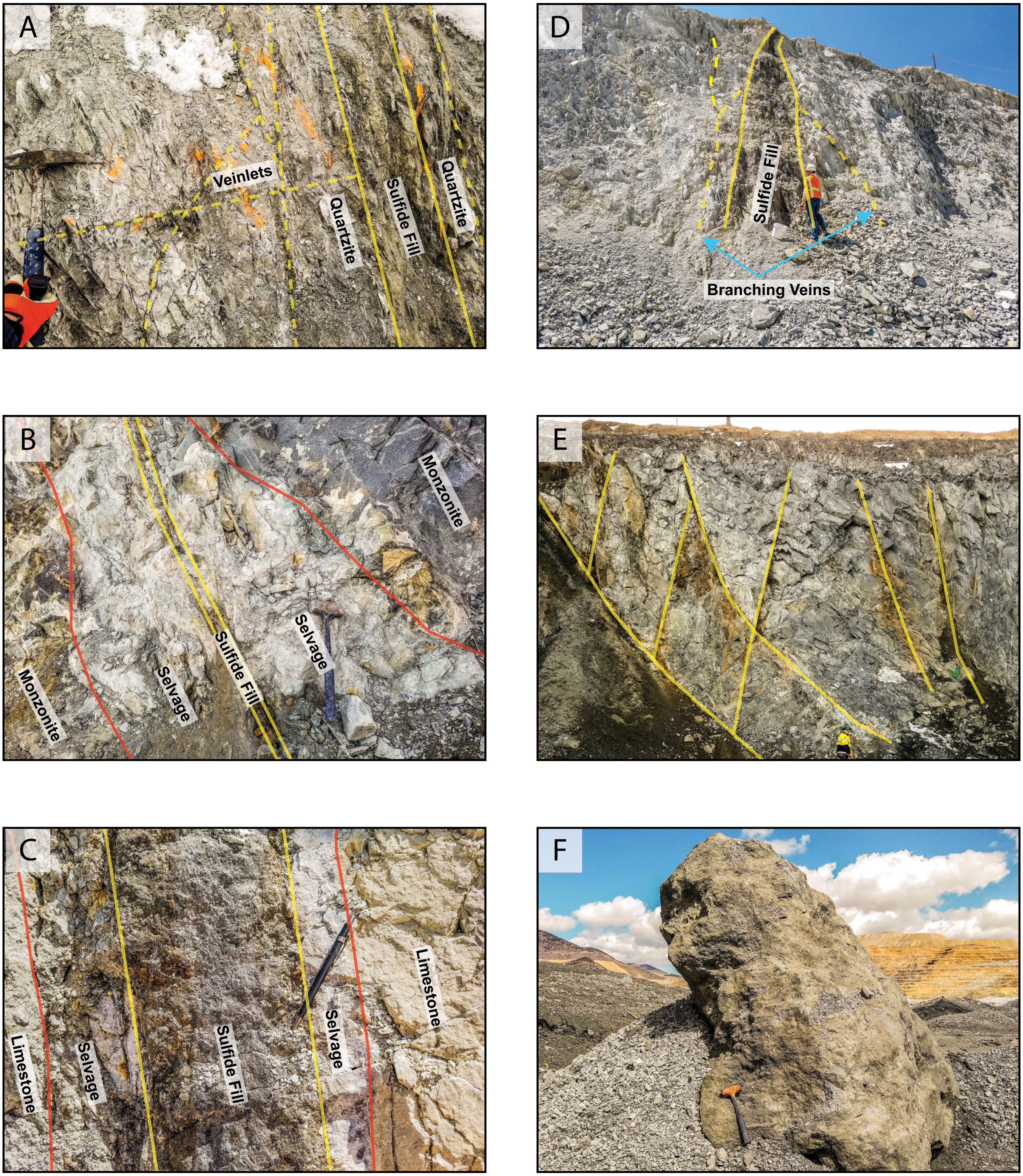

Fig. 4. Macroscale features of polymetallic vein mineralization at the Bingham deposit. A. Intense brecciation and fracturing in a quartzite-hosted vein with small pyrite and quartz veinlets that are both perpendicular and parallel to the vein. B. Wide mixed quartz and sericite selvages bounding a monzonite-hosted vein that are weak, friable, and lighter in color relative to the darker unaltered wall rock. C. Zoned limestone-hosted vein with galena, pyrite, chalcopyrite, and sphalerite in the middle with predominately pyrite toward the edges followed by a light-colored silicified selvage. D. A large vein that splays into a few smaller branching veins. E. Example of a vein "swarm" in propylitically altered monzonite. Such swarms are observed in all rock types but particularly in quartzite. F. A fine-grained sulfide block separated from a vein with interstitial quartz shows how large and competent these mineralized bodies can be.

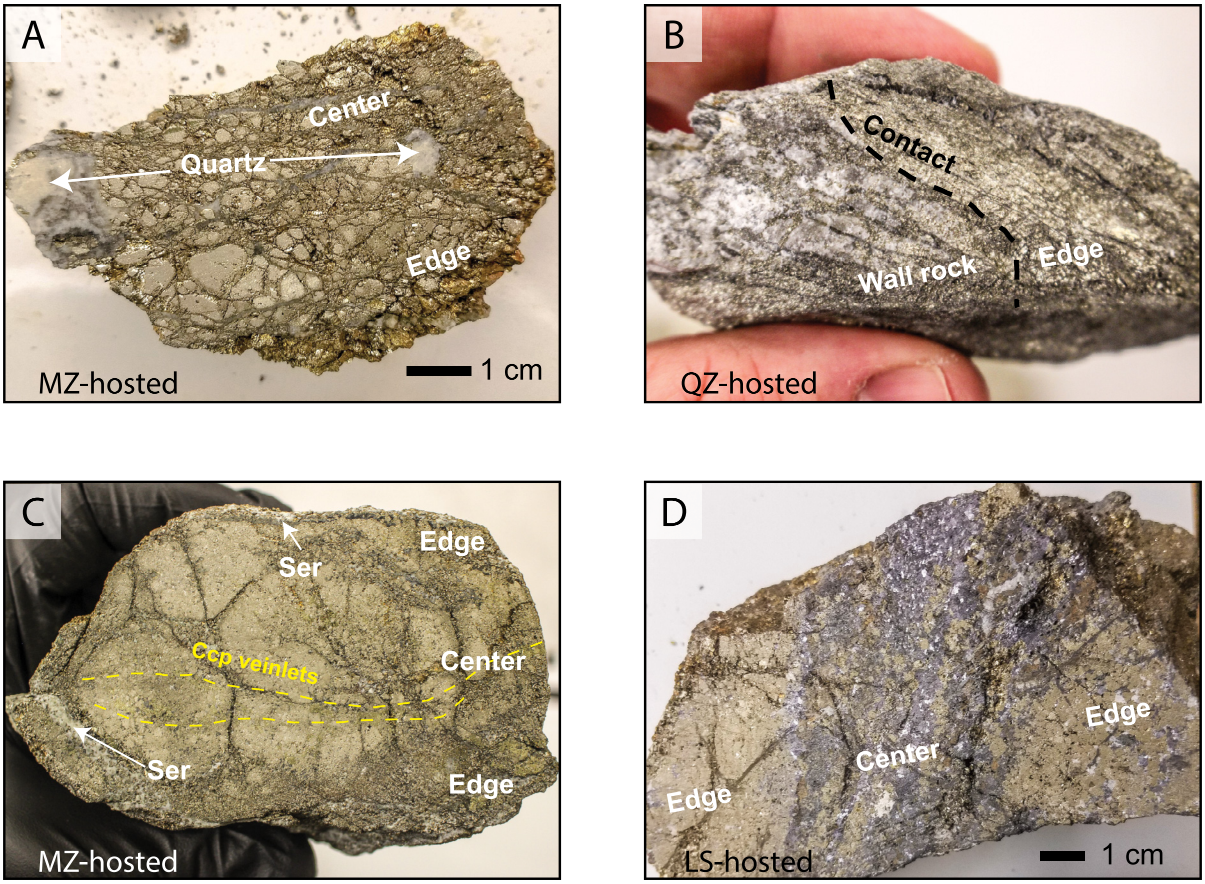

Fig. 5. Slabs cut from the fill material of various polymetallic veins. A. Monzonite-hosted vein from the Fe-Cu zone with quartz-filled vugs and fractures. B. Contact between a vein and quartzite in the Fe-Cu zone. Small sulfide veinlets extend into the quartzite. C. Vein from the more distal Pb-Zn-Cu-Ag-Au zone with sericite (Ser) selvage where it meets monzonite wall rock and small chalcopyrite (Ccp) veinlets that parallel the main vein orientation. D. Limestone (LS)-hosted vein from the Pb-Zn-Cu-Ag-Au zone with pyrite-dominated edges and abundant galena and sphalerite toward the center.

occurs in polymetallic veins from other porphyry deposits (Muntean and Einaudi, 2001; Einaudi et al., 2003; Sillitoe, 2010; Maydagán et al., 2013).

Galena and sphalerite are restricted to the polymetallic veins farthest from the center of the deposit (FF2, FF3, and FF4; Figs. 5D, 6D). A crude Pb-Zn boundary was drawn in Figure 2 to illustrate where this occurs and was further extrapolated based on Pb and Zn assays. These were the primary ore minerals for the early underground workings on the southern outer edges of the current pit. North of this boundary, galena and sphalerite are not found in the veins either as inclusions or as separate grains (Table 2). South of the Pb-Zn boundary veins also contain the most chalcopyrite. Tennantite generally appears in the veins at this same boundary, though minor tennantite was found in a monzonite-hosted vein (FF5) slightly closer to the center of the deposit.

Microscopic physical characteristics of polymetallic veins variable by host rock

Quartzite-hosted veins contain the simplest mineral assemblages consisting of pyrite and quartz with minor sericite and chalcopyrite. Mineral inclusions in the pyrite include quartz, chalcopyrite, apatite, and anhydrite. Brecciation is most intense in the quartzites and produces microfaults and complete shattering of individual grains that are not as readily apparent in other rock hosts (Fig. 6F). The effects of hydrothermal alteration on the quartzite are minimal and manifested primarily by secondary quartz, sericite, calcite, and sulfides in veinlets and in some interstitial space. The veinlets show a transition in mineral proportions with distance from the main vein. They begin with pyrite > quartz > sericite within 0 to 1.5 m, transitioning to quartz > sericite > pyrite within 1.5 to 3.0 m, and ending with sericite > quartz at 6 m.

Monzonite-hosted veins have pyrite as the dominant sulfide with minor chalcopyrite and tennantite. The dominant interstitial gangue mineral varies from sericite (FF2), to calcite (FF5), or quartz (FF7). In addition to multiple generations of quartz, calcite and sericite appear to have formed in two generations (one filling interstitial space and the other healing fractures), which cut the first and second generations of quartz and sulfides. These veins have coarser sulfide and gangue mineral grains than in other rock types. The largest grains are consistently in the center of the vein and grade outward to more fine-grained material on the edges (Fig. 6G). This reduction in grain size is accompanied by an increasing abundance of gangue minerals (typically sericite). Vein-related wall-rock alteration is essentially absent 0.3 to 1.5 m away, but small pyrite-quartz-sericite veinlets can continue

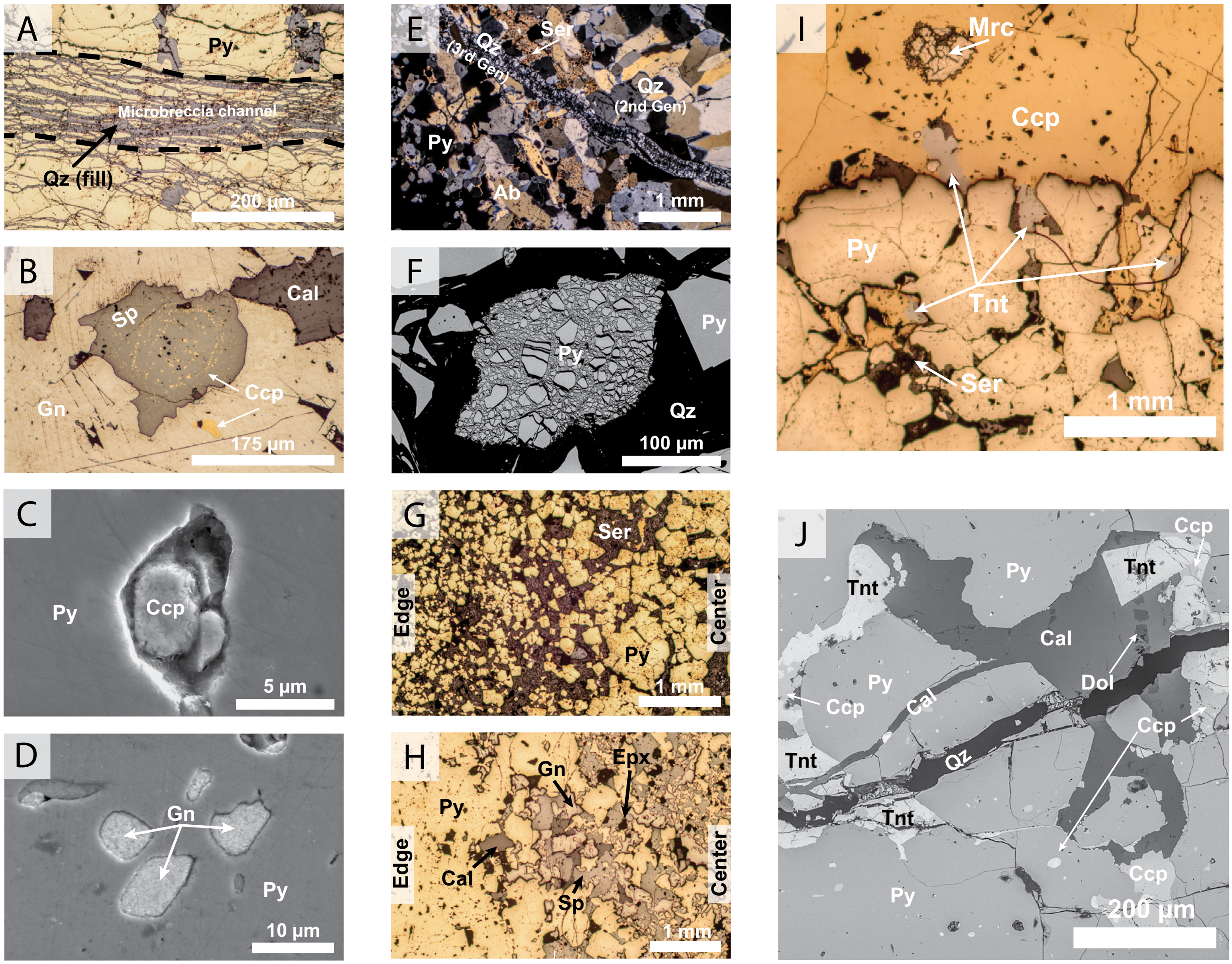

Fig. 6. Microscale features of the sulfide fill of polymetallic veins at the Bingham deposit. A. Microbreccia "channels" that are present in all rock types and are parallel to the associated veins (reflected light). B. Sphalerite grain from the Pb-Zn-Cu-Ag-Au zone with chalcopyrite inclusions that form rings that mimic growth zones (reflected light). C. Typical chalcopyrite inclusion in a pyrite grain (SEM image). D. Galena inclusions in a pyrite grain that only appear outside the Pb-Zn boundary shown in Figure 2 (SEM image). E. Quartz microveinlet cutting across an earlier generation of quartz, sericite, albite, and pyrite (cross-polarized transmitted light) F. Shattered pyrite grain from a faulted vein (SEM image). G. Fining outward trend of pyrite grains in a monzonite-hosted vein with increasing amounts of interstitial sericite as the altered wall rock is approached (reflected light). H. Fining inward of sulfides in a limestone-hosted vein, which is also associated with more abundant pyrite on the edges and galena, sphalerite, calcite, and pyrite in the center (Fig. 4C; reflected light). I. Chalcopyrite veinlet next to pyrite in a monzonite-hosted vein with chalcopyrite and tennantite textures indicating they are late stage as the crystals are typically anhedral filling interstitial space and healed fractures (reflected light). J. Another example of the association and textures of chalcopyrite and tennantite, which are found close to a small third-generation quartz veinlet (BSE image). Ab = albite, Cal = calcite, Ccp = chalcopyrite, Dol = dolomite, Epx = epoxy, Gn = galena, Mrc = marcasite, Py = pyrite, Ser = sericite, Sp = sphalerite, Tnt = tennantite.

past 12 m depending on the width of the main vein. Biotite grains proximal to these veinlets are chloritized. These veinlets change in mineral proportions with increasing distance from main vein with the sulfides disappearing first leaving predominantly sericite in the veinlets. The abundance of sericite in the monzonites, whether in the veins or altered wall rock, is significantly greater than in the other rock types.

Limestone-hosted veins also have pyrite as the dominant sulfide, although they can also have significant galena and sphalerite with minor chalcopyrite, which increase with depth in the limestone-hosted polymetallic veins (Field and Moore, 1971) and nearer to the center of the deposit. Chalcopyrite is present both as anhedral grains and as inclusions in sphalerite (Fig. 6B). Interstitial minerals are calcite and dolomite with lesser amounts of quartz and sericite. Unlike veins in other rock types, limestone-hosted veins can have large galena and sphalerite grains that are similar in size to the pyrite. These veins are also zoned with large pyrite grains on the outer edges that fine inward toward the center of the fill material (Fig. 5D). The central fill is intergrown, fine-grained, anhedral

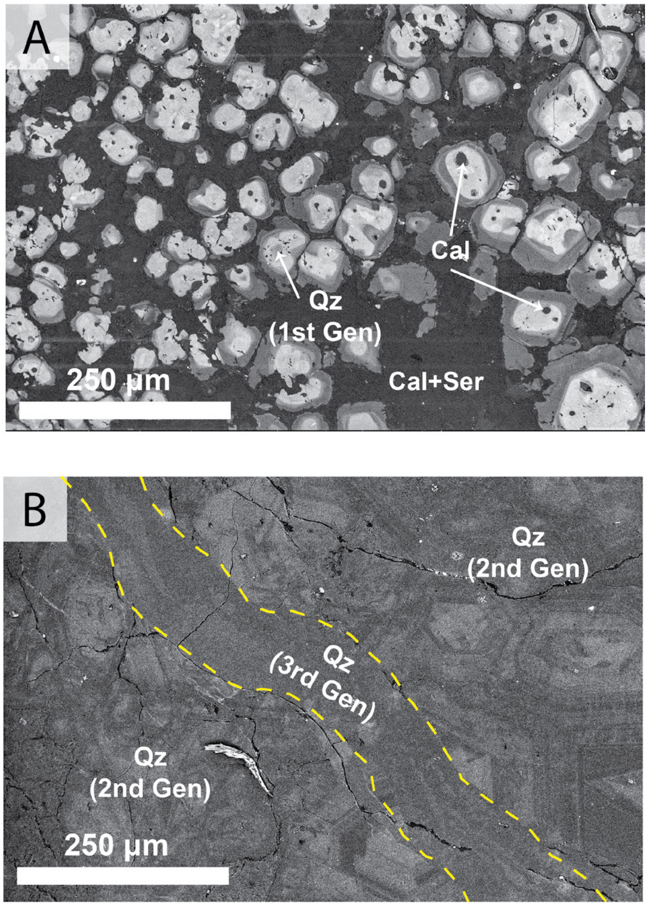

Fig. 7. SEM-CL images of three generations of quartz grains in the Bingham polymetallic veins. A. The first generation of quartz, which is identified based on CL brightness and textures, is found near vein margins. Rounded and embayed cores (bright) are assumed to be from hotter hydrothermal fluids followed by dissolution. These were overgrown by lower temperature quartz (dark) with more euhedral faces. B. The second and third generations of quartz are identified based on crosscutting relationships and CL brightness (see Fig. 6E). The second generation is also bright with oscillatory zoning and large euhedral grains, while the third generation has a dull luminescence and is finer grained.

Table 3. Select Whole-Rock Chemical Analysis of Polymetallic Veins

		Fe-Cu			Pb-Zn-Cu-Ag-Au		
Host rock		QZ	MZ	MZ	MZ	LS	QZ
Vein		FF1-C	FF5-C	FF7-C	FF2-C	FF3-C	FF4-C
Ag	ppm	1.30	1.30	1.00	76.8	511	6.60
As	ppm	52	69	65	1,855	3,510	380
Au	ppm	0.27	0.67	0.94	4.61	2.24	1.03
Ba	ppm	5.0	0.5	0.5	50.0	0.5	0.5
Bi	ppm	10	20	10	115	389	16
Cd	ppm	0.3	0.9	1	2.5	274	0.8
Co	ppm	21	43	23	44	8	19
Cr	ppm	0.5	1	0.5	42	0.5	3.0
Cu	ppm	66	1,450	112	23,000	8,430	113
Ga	ppm	0.5	1	0.5	20.0	0.5	0.5
La	ppm	0.5	1	0.5	10	0.5	20.0
Mn	ppm	2.5	277	2.5	518	313.0	2.5
Mo	ppm	0.5	1	1	3	1	1
Ni	ppm	3	5	4	20	2	2
P	ppm	5.0	20	5	1,310	10	10
Pb	ppm	135.0	51	190.0	853.0	>200,000	681.0
Sb	ppm	2.5	7	12	30	513	14.0
Sc	ppm	1	1	1	5	1	1
Sr	ppm	0.5	8.0	0.5	12.0	1.0	2.0
Th	ppm	10	10	10.0	10.0	10.0	10
Tl	ppm	1	1	1	1	0.5	1
U	ppm	20	1	10	30	1	1
V	ppm	1	1	1	49	1	4
Zn	ppm	20	12	183	246	57,400	317
Al_2O_3	wt %	0.13	0.11	0.08	4.65	0.04	0.74
Fe_2O_3	wt %	56.33	58.91	51.47	50.61	36.74	50.90
CaO	wt %	0.11	3.71	0.01	2.43	1.26	0.04
MgO	wt %	0.02	0.02	0.01	1.53	0.02	0.02
Na_2O	wt %	0.01	0.01	0.01	0.01	0.01	0.01
K_2O	wt %	0.04	0.04	0.02	1.13	0.01	0.20
TiO_2	wt %	0.01	0.01	0.01	0.22	0.01	0.01

Notes: Detection limits vary between elements due to the variety of analytical methods used (see Methods); values below detection limits for trace elements (typically 1 ppm) were reported as half of the detection limit; silica was not analyzed in the vein material as it volatizes during acid digestion

pyrite, galena, and sphalerite in roughly equal proportions (Fig. 6H). Contacts between the polymetallic veins and limestone wall rock are generally sharp. Selvages are dominated by complete silicification of the limestone, which gradually fades out within <1 m. Silicified rock is cut by narrow sericite-sulfide-carbonate veinlets. Wall-rock veinlets greater than about 3 m from the main vein contain just sericite and carbonate minerals.

Chemical Characteristics of Polymetallic Veins

Whole-rock geochemistry of polymetallic veins

The sulfide-rich fill material of each vein contains greater Ag, As, Au, Bi, Cd, Cu, Fe, Pb, Sb, and Zn concentrations than associated wall rocks (Table 3). Tin was not analyzed in the vein fill, but the wall-rock concentrations of Sn all increase toward the veins. Gold concentrations are weakly correlated with Ag, As, and Cu (Fig. 8), which suggests that Au may be hosted in a mineral phase rich in those elements, such as tennantite. Most of these elements (excluding Fe) also increase with increasing distance from the center of the deposit in the veins and show weak positive correlations with each other. The strongest correlations with distance are with Ag, As, Bi, Pb, and Zn, which are highest in limestone-hosted veins. The concentrations of these elements can be several orders of magnitude greater in limestone-hosted veins than in the other host rocks (Table 3).

The sharpest increase in all of these enriched elements (Ag, As, Au, Bi, Cd, Cu, Pb, Sb, and Zn) occurs at the Pb-Zn boundary (Fig. 2), which is accompanied by a shift in metal ratios (Table 4). The average Ag/Au ratio of the vein fill sampled inside the Pb-Zn boundary (Fig. 2) is 16/1, whereas the average beyond the boundary is 79/1. High Ag/Au ratios have been suggested as a characteristic of an intermediate-sulfidation epithermal environment (Einaudi et al., 2003), although John et al. (2018) noted this is not necessarily diagnostic as there are intermediate-sulfidation deposits with no reported Ag. The Cu/Au ratios also show a strong contrast between the two zones. Polymetallic veins and associated wall-rock samples in the inner Fe-Cu zone have an average Cu/Au ratio of

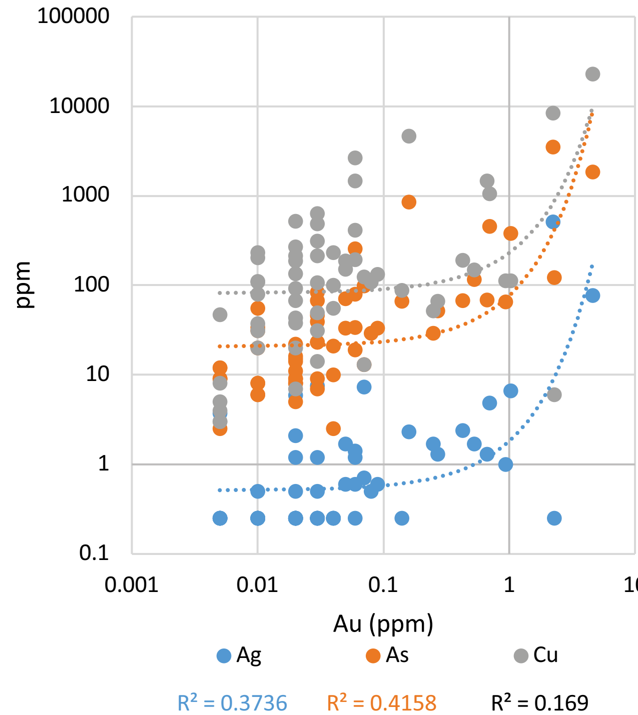

Fig. 8. Logarithmic plot of Ag, As, and Cu vs. Au concentrations in whole-rock samples of vein fill material and associated wall rocks up to 6 m from the edge of each vein. All of these elements show weak positive correlations with Au, the strongest of which is with As. Dotted lines are exponential fit lines.

10,300/1 and those in the outer Pb-Zn-Cu-Ag-Au zone have an average ratio of only 2,333/1 (Fig. 2).

Whole-rock geochemistry of wall rocks

Isocon diagrams are constructed by comparing whole-rock analyses of altered wall rock against fresh wall rock (Grant, 1986). The slopes of the isocons reflect dilution (m <1) or residual enrichment (m >1) of the immobile elements. For our purposes, we considered Zr and Ti immobile elements during hydrothermal alteration and the isocon was drawn by fitting a line to those two elements. Elements above the isocon indicate enrichment, whereas those below indicate depletion. We show four such diagrams of monzonite wall-rock samples near vein FF2 compared to a monzonite sample unaffected by vein-related alteration (MZ-2) in Figure 9. Samples closest to the vein are enriched in ore metals (Ag, Au, and Cu), as well as As, Cd, Cr, Fe, Ni, Pb, Sn, and Zn, and importantly Si. They are depleted in mobile alkalis (Na, K, Rb, and Cs) and alkaline earths (Ba, Sr). The slope of the isocon also reveals that the concentrations of the typically immobile elements during hydrothermal alteration were strongly diluted by the introduction of silica near the veins (Fig. 9). For example, the high silica content of the sample at 0.3 m (72 vs. 56 wt % in the unaltered equivalent) diluted the immobile elements causing a shallow slope for the isocon. Silicification is not obvious at 1.5 to 6.0 m; silica lies near the isocon and the slopes of the isocons are nearly 1/1 showing element mobility was very limited.

The wall rocks essentially return to their original compositions except for the most mobile elements (Na is depleted and Au, Ag, Mo, and Pb are enriched) greater than 1.5 m away from the veins, which is consistent with megascopic effects of alteration and mineralization on the wall rock. The only exception was for an abnormally wide (~1.5 m) vein (FF7, Fig. 4D), which had abundant sulfides along joints up to 12 m away. The restricted lateral effects on the wall rocks are true for veins regardless of rock type or proximity to the center of the deposit.

Chemical characteristics of sulfides in polymetallic veins

Pyrite: Pyrite was studied in detail, because it is ubiquitous in each vein regardless of location or host rock. Other sulfide phases were also investigated in less detail. *Pyrite* trace element compositions parallel many of those noted in the whole-rock geochemistry of the vein fill because they are the largest constituent. Those analyzed in the Pb-Zn-Cu-Ag-Au zone are enriched in many trace elements relative to those in the Fe-Cu zone. This enrichment is particularly evident for Ag, As, Au, Bi, Cd, Cu, Ni, Pb, Sb, Sn, and Zn (Table 5). Similar trace element enrichments have been found in pyrite from epithermal deposits associated with other porphyry Cu systems (Maydagán et al., 2013; Franchini et al., 2015; Sykora et al., 2018).

Core and rim trace element LA-ICP-MS analyses on pyrite in the outer Pb-Zn-Cu-Ag-Au zone (veins FF2 and FF3) reveal considerably higher concentrations of chalcophile trace elements than pyrite formed in the Fe-Cu zone (Fig. 10). If we consider only the grains that showed the highest concentration of trace elements, some generalizations can be made: Ag, Cd, Cu, Ni, Sb, Sn, and Zn are typically more enriched in the cores of pyrite in the Pb-Zn-Cu-Ag-Au zone, whereas As and Au are typically more enriched in the rims.

Concentrations of Cu and As range from sub-ppm to several 1,000 ppm in the Pb-Zn-Cu-Ag-Au pyrite zone. They show a negative correlation when either one reaches concentrations in the 1,000s of ppm. The decoupling of Cu and As in pyrite has been observed in several types of deposits (Tanner et al., 2016; Tardani et al., 2017), including porphyry Cu deposits (Reich et al., 2013), and is associated with changes in

Table 4. Average Metal Ratios of Polymetallic Vein Ore

	Fe-Cu				Pb-Zn-Cu-Ag-Au			
Host rock	QZ	MZ	MZ		MZ	LS	QZ	
Vein	FF1	FF5	FF7	Average	FF2	FF3	FF4	Average
Ag/Au	29	11	10	16	25	199	14	79
Cu/Au	10,600	9,200	11,100	10,300	2,900	2,500	1,600	2,333

Notes: Average metal ratios are determined from combined whole-rock analyses of veins and associated wall-rock samples for each sample location

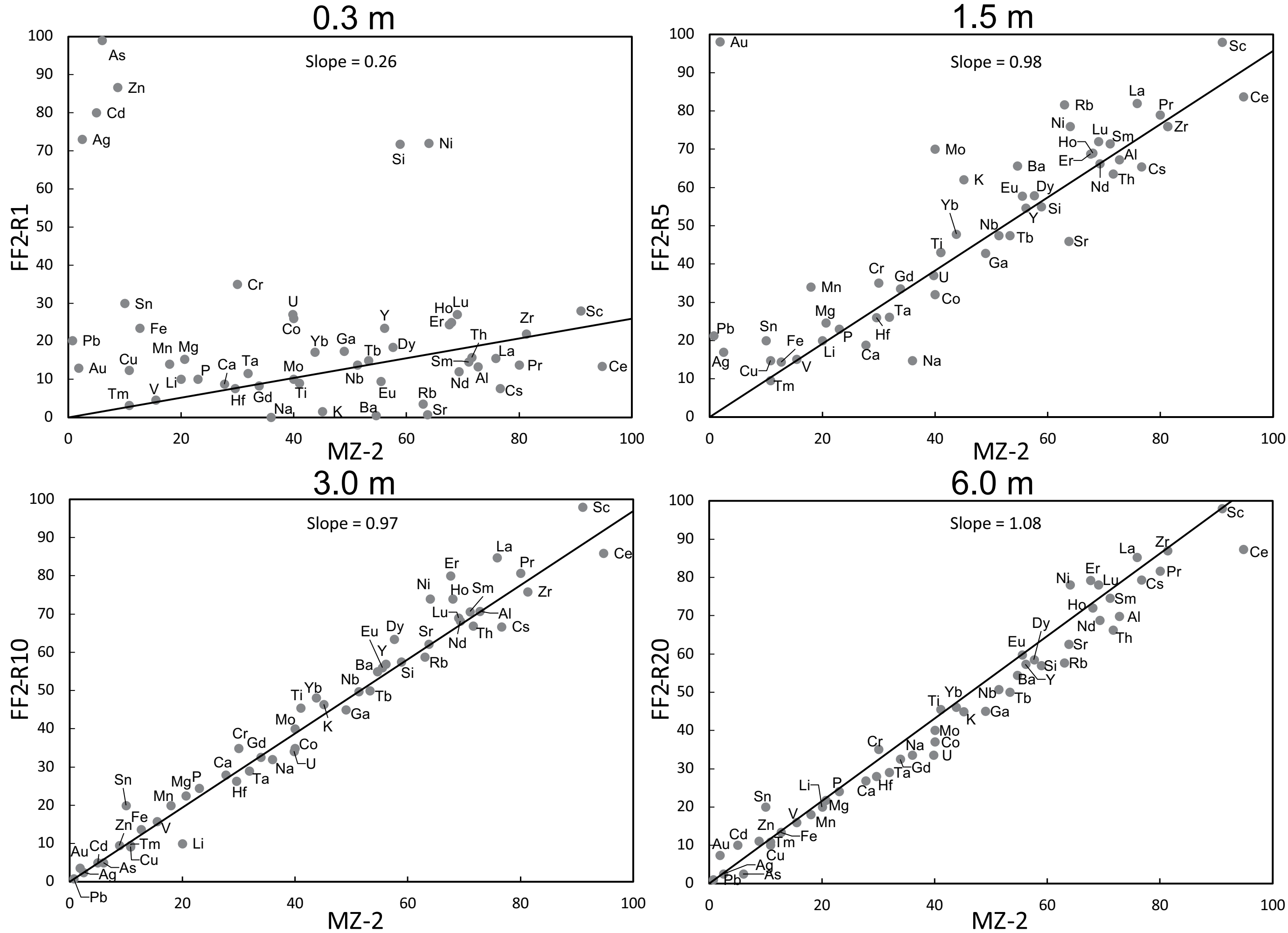

Fig. 9. Isocon diagrams for wall rock at intervals away from a polymetallic vein (FF2). A propylitically altered monzonite sample that was geochemically unaffected by vein mineralization was chosen to represent the original composition (MZ-2). Element concentrations are arbitrarily scaled for clarity. The black line represents the isocon, which was drawn by fitting a line to Ti and Zr. Both of these elements are typically considered immobile during hydrothermal alteration (Grant, 2005). Other elements that consistently fall near this line may also be considered immobile. Elements above the isocon indicate enrichment, whereas those below indicate depletion.

the composition of the mineralizing fluids (Reich et al., 2013; Tardani et al., 2017). Pačevski et al. (2008) determined that as much as 8 wt % Cu can be structurally bound in hydrothermal pyrite; here, Cu concentrations are much lower and reach only as much as 0.31 wt %. Recent observations have revealed the significance of nano-sized (<1 μm) chalcopyrite inclusions in pyrite, which contribute to Cu enrichment, in addition to solid solution (Reich et al., 2013; Tardani et al., 2017). We saw no such evidence of chalcopyrite inclusions during our analysis of "clean" pyrite areas, leading us to believe the Cu from our analyses is structurally bound.

In some ore deposits, gold forms as nanoparticles in As-rich rims of pyrite, especially in Carlin-type deposits (Palenik et al., 2004; Reich et al., 2005). Reich et al. (2005) determined that in Carlin-type and epithermal deposits, the Au in pyrite only occurs as nanoparticles, as opposed to solid solution, when the Au/As ratio is greater than 0.02. The average Au/As ratio from Au-bearing pyrite in our samples is significantly lower—0.001. This suggests that Au is likely present as solid solution rather than as nanoparticles in the vein pyrite. High Au concentrations have also been found in As-free pyrite in the form of telluride inclusions in other deposits (Cook et al., 2009a), but the single telluride microinclusion that we found did not contain detectable concentrations of Au (>2 wt %) based on SEM-EDS analyses.

Despite the presence of Au in the pyrite at the ppm level, it alone cannot be the source of Au in the veins based on simple mass-balance calculations. The highest whole-rock Au concentration from our sample set was 4.6 ppm, whereas the highest Au concentration in the pyrite from that same sample was only 1.0 ppm. The modal percent of pyrite based on visual observations in the sample was approximately 80%, which is fairly representative of most of the veins. Even if we assume that every pyrite grain had 1.0 ppm Au, it could only account for 18% of the total Au in the rock. This indicates that another mineral phase or phases supplies the majority of the Au in the ore with several 10s of ppm Au. Alternatively, our selection for laser ablation analyses of "clean" locations free of inclusions

Table 5. Select Major and Trace Element Compositions of Pyrite from Vein Fill

		Fe-Cu						Pb-Zn-Cu-Ag-Au					
Grain		FF1-3	FF1-3	FF5-3	FF5-3	FF7-1	FF7-1	FF2-1	FF2-1	FF3-3	FF3-3	FF4-2	FF4-2
Host rock		QZ	QZ	MZ	MZ	MZ	MZ	MZ	MZ	LS	LS	QZ	QZ
Location		Core	Rim	Core	Rim	Core	Rim	Core	Rim	Core	Rim	Core	Rim
Fe	wt %	46.20	46.53	46.90	46.13	46.71	46.39	45.45	45.68	46.31	46.38	46.60	46.36
S	wt %	54.01	54.07	52.98	53.45	53.45	53.59	54.05	53.82	53.35	52.84	54.09	53.69
Total		100.21	100.60	99.88	99.58	100.15	99.99	99.50	99.50	99.66	99.22	100.69	100.05
Ag	ppm	mdl	mdl	mdl	mdl	mdl	mdl	16	13	mdl	2.8	mdl	mdl
As	ppm	27	1.9	mdl	mdl	mdl	mdl	270	7100	100	4200	8.4	2.8
Au	ppm	mdl	mdl	mdl	mdl	mdl	mdl	0.8	1.0	mdl	1.5	mdl	mdl
Bi	ppm	3	1.2	mdl	0.3	mdl	1.0	48	48	0.1	0.4	mdl	0.4
Co	ppm	1.7	16	4.2	9.7	1.4	0.7	mdl	mdl	mdl	2.2	0.4	0.6
Cr	ppm	2	mdl	mdl	5	mdl	3	mdl	mdl	2	3	mdl	mdl
Cu	ppm	5	28	20	2.7	73	62	2000	140	2.0	23	mdl	40
Mn	ppm	1.8	2.1	2.3	2.3	2.3	2.9	4.1	mdl	1.7	2.0	1.4	1.7
Ni	ppm	mdl	6	1.2	8	mdl	2.1	mdl	mdl	mdl	5	mdl	mdl
Pb	ppm	24	1.7	mdl	mdl	mdl	1.3	30	170	4	52	0.2	2
Sb	ppm	mdl	mdl	mdl	mdl	mdl	mdl	2.4	mdl	mdl	4.9	mdl	mdl
Se	ppm	36	30	70	60	170	100	80	40	3	12	mdl	5
Sn	ppm	0.7	4	0.6	0.4	mdl	0.7	60	1.0	mdl	0.8	mdl	mdl
Te	ppm	50	0.9	mdl	mdl	mdl	4	mdl	13	mdl	2.8	0.2	2.3
Zn	ppm	0.6	0.5	mdl	1.4	mdl	1.8	2270	3	1.6	0.4	mdl	mdl

Notes: Major elements (Fe, S) were determined by EMPA and remaining elements were determined by LA-ICP-MS; concentrations below detection limits are reported as mdl

and cracks may have introduced an inherent bias. Gold has been found preferentially located along healed fractures or in microinclusions in some Carlin-type and orogenic gold deposits (Cook et al., 2009a; Large et al., 2009), which would have been missed with our analytical methods.

Chalcopyrite: Chalcopyrite typically had Au concentrations below detection limits (~0.4 ppm) apart from one grain with 3.1 ppm Au (Table 6). This grain also had anomalously high Ag, As, and Pb, suggesting a possible mineral inclusion that was not observed during grain inspections. Ignoring that

Table 6. Select Major and Trace Element Compositions of Chalcopyrite from Vein Fill

		Spot analyses					Veinlet traverse				
Grain		FF2-CPY2	FF2-CPY3A	FF3-CPY1	FF5-CPY3A	FF5-CPY4	FF2-CPY4A	FF2-CPY4B	FF2-CPY4C	FF2-CPY4D	FF2-CPY4E
Host rock		MZ	MZ	LS	MZ	MZ	MZ	MZ	MZ	MZ	MZ
Cu	wt %	34.19	33.79	34.04	34.77	33.90	33.91	34.34	34.18	34.41	34.40
Fe	wt %	30.32	30.44	30.21	27.71	30.54	30.53	30.66	30.35	30.38	30.25
S	wt %	35.13	34.79	35.09	35.01	34.64	35.29	35.36	35.09	34.97	34.86
Total		99.64	99.02	99.34	97.49	99.08	99.73	100.36	99.61	99.76	99.51
Ag	ppm	3.8	13.7	mdl	980	mdl	8.4	9.3	35	9.5	11.8
As	ppm	mdl	15	mdl	8400	mdl	4	mdl	11	mdl	mdl
Au	ppm	mdl	mdl	mdl	3.1	mdl	mdl	mdl	mdl	mdl	mdl
Bi	ppm	0.4	1.7	mdl	1.9	0.5	0.6	0.4	mdl	mdl	1.3
Cd	ppm	12	mdl	mdl	mdl	mdl	mdl	20	22	17	12
Cr	ppm	mdl	30	5	9	8	mdl	mdl	11	mdl	3
Ga	ppm	mdl	7	mdl	mdl	mdl	mdl	mdl	mdl	5	9
Hg	ppm	mdl	mdl	mdl	1.5	mdl	mdl	mdl	mdl	mdl	mdl
In	ppm	46	87	108	mdl	1.9	76	69	64	64	61
Mn	ppm	23	mdl	14	9	mdl	14	18	32	39	12
Pb	ppm	1.7	3.4	6	130	mdl	1.2	1.0	3.0	1.0	1.0
Sb	ppm	mdl	8	mdl	8000	mdl	mdl	mdl	4	mdl	mdl
Se	ppm	50	20	20	40	130	30	10	40	10	50
Sn	ppm	97	400	330	2.0	380	219	128	105	270	560
Tl	ppm	mdl	mdl	mdl	73	mdl	mdl	mdl	mdl	mdl	mdl
Zn	ppm	4000	530	170	49	8	610	710	1600	660	560

Notes: Major elements (Cu, Fe, S) were determined by EMPA and remaining elements were determined by LA-ICP-MS; concentrations below detection limits are reported as mdl

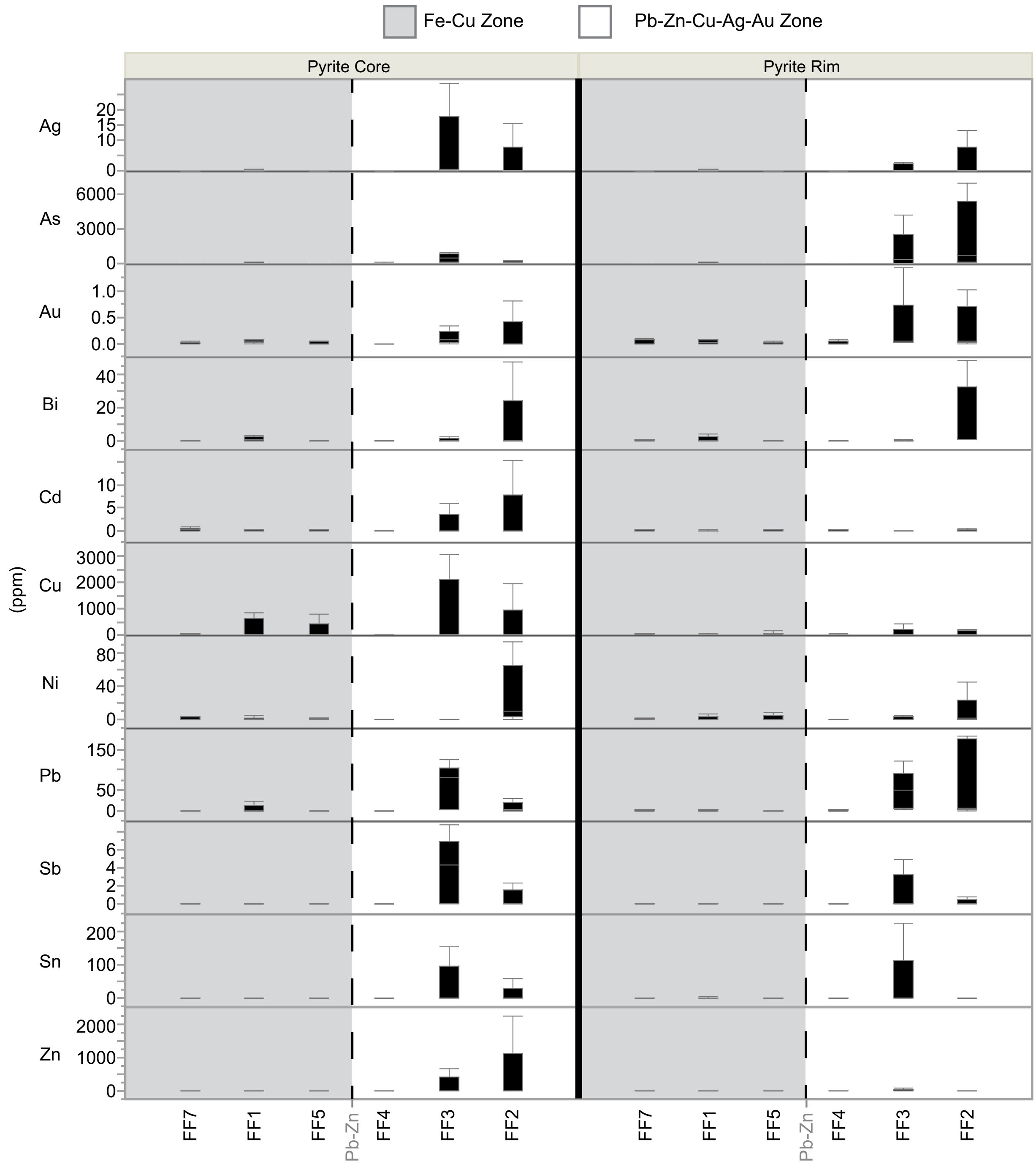

Fig. 10. Box and whisker plots of trace element concentrations in pyrite grains ($n = 60$) from both sides of the Pb-Zn boundary in Figure 2, which is illustrated here by the gray dashed line. The bottom and top of each plotted box reflects the 1st and 3rd quartiles, respectively. Values outside of the interquartile range are plotted as whiskers. Polymetallic veins are ordered from left to right by increasing distance from the center of the deposit. Pyrite grains in the outer Pb-Zn-Cu-Ag-Au zone (FF3 and FF2) are enriched in these trace elements relative to those from the Fe-Cu zone (FF7, FF1, FF5).

outlier gives ore metal concentrations of 33.6 to 34.9 wt % Cu, 3.8 to 45 ppm Ag, and undetectable amounts of Au. George et al. (2018) found that hydrothermal chalcopyrite is not a preferential host for Au with maximum concentrations usually less than a few ppm regardless of deposit type. The chalcopyrite compositions in the Bingham polymetallic veins support this conclusion. The chalcopyrite does contain appreciable amounts of In, Mn, Se, Sn, and Zn with lesser Ag, As, Bi, Cd, Cr, Ga, Pb, and Sb (Table 6).

It is evident that the chalcopyrite is late stage, but we can further constrain how the fluids evolved toward the waning stages by examining a traverse across a chalcopyrite veinlet from the center of vein FF2. Five equally spaced spots were analyzed (Table 6; FF2-CPY4A–FF2-CPY4E) across a 3-mm chalcopyrite veinlet in FF2 with the first and last analysis representing the edges of the veinlet. Bismuth, Ga, In, and Sn concentrations all decrease toward the center of the veinlet, whereas Ag, As, Cd, Cr, Mn, Pb, Sb, Se, and Zn all increase. These trends are replicated on both sides of the veinlet often with a two-fold difference between the edge and center concentrations.

The Cd/Zn ratios in coprecipitated chalcopyrite and sphalerite vary together and should remain roughly constant (George et al., 2018). Chalcopyrite from the fissures has an average Cd/Zn ratio of 0.02, whereas sphalerite has an average ratio of 0.006. This suggests that they did not coprecipitate, which is supported by chalcopyrite textures in thin sections indicating it formed later than the sphalerite. Chalcopyrite microinclusions in sphalerite, such as those shown in Figure 6B, were too small for LA-ICP-MS analysis.

Sphalerite: Sphalerite grains are all strongly enriched in Mn and Cd (1,000s ppm), with moderate Ag, In, and Sn (10s–100s ppm), and with minor amounts of Cr, Ga, As, Se, Sb, Au, Hg, and Pb (<10–10s ppm; Table 7). The average Cu concentration is ~8,500 ppm, which is higher than the galena (max 140 ppm) or pyrite (max 3,100 ppm) and may be influenced by subsurface chalcopyrite inclusions not observed during grain inspections. However, sphalerite in other deposits is enriched in the same elements, particularly Mn, Cu, and Cd, which are consistently in the 1,000s ppm (Palero-Fernández and Martín-Izard, 2005; Cook et al., 2009b). The preferential partitioning of Ga in sphalerite relative to the chalcopyrite is typical for epithermal conditions (George et al., 2016). The average FeS content in the grains was 3.1 ± 0.5 mol % (Table 7), which aligns with sphalerite found in low- to intermediate-sulfidation epithermal deposits (John et al., 2018). The sphalerite grains do not appear to be homogeneous in trace elements with concentrations between different spots on the same grain varying two to three times in several cases suggesting significant trace element zonation, which occurs in other deposits as well (George et al., 2016). Sphalerite in vein FF3 contains appreciable Cu (3,400–15,300 ppm), but significantly less Ag (16–43 ppm) and Au (≤0.2–0.3 ppm). Ragged ablation spectra for Ag, Pb, and Sb in sphalerite from other deposits have been attributed to nanoscale inclusions of sulfosalts (Cook et al., 2009b). In the sphalerite from FF3, Pb and Sb spectra show this raggedness to an extent, although both had typically low concentrations (avg 25 ppm Pb and 2.5 ppm Sb) and so the variation could be analytical. Silver did not have such ragged signals, which leads us to believe the Ag concentrations reported here are representative of solid solution. This cannot be determined as confidently for Au, but the concentration (when detectable) was limited to a restricted range among different grains, which supports Au being in solid solution in the sphalerite as well.

Galena: Galena in vein FF3 was most notably enriched in Ag, Bi, and Sb, with moderate amounts of As, Cd, Cu, Fe, Tl, and Zn, and lesser amounts of Au, Cr, In, Mn, Se, Sn and Te (Table 8). Like sphalerite, galena shows enrichments of chalcophile trace elements similar to those in several deposit types (George et al., 2015). Comparison between the cocrystallized galena and sphalerite in the veins shows a clear partitioning of trace elements. Cadmium, Cu, Ga, Hg, In, Mn, Sb, and Sn show a preference for sphalerite, whereas Ag, As, Bi, Sb, Te, and Tl show a preference for galena. Concentrations of Ag in galena from vein FF3 range from 560 to 1,220 ppm, but Cu and Au were typically below detection limits. The sulfide fill of vein FF3 had 511 ppm Ag and the average concentration of Ag in galena from that sample was 955 ppm. All the minerals can contribute to the Ag content to an extent; however, galena alone can contribute all of the Ag if the modal abundance is 52%, which is a reasonable amount (Fig. 5D).

Tennantite: Tennantite ($Cu_{12}As_4S_{13}$ or $Cu_6[Cu_4(Fe,Zn)_2]As_4S_{13}$) is the most compositionally complex mineral analyzed.

Table 7. Major and Trace Element Compositions of Sphalerite from Vein FF3

Grain Host rock		FF3-SPH1A LS	FF3-SPH1B LS	FF3-SPH2A LS	FF3-SPH2B LS	FF3-SPH3A LS	FF3-SPH3B LS	FF3-SPH4A LS	FF3-SPH4B LS
Zn	wt %	64.68	64.20	63.42	63.51	59.77	59.75	65.05	64.94
Fe	wt %	2.15	2.11	1.57	1.71	2.38	2.24	2.19	2.04
S	wt %	33.29	33.24	33.28	33.43	33.05	32.98	33.15	33.32
Total		100.12	99.55	98.27	98.65	95.20	94.97	100.39	100.30
FeS	mole %	3.22	3.18	2.42	2.62	3.83	3.61	3.26	3.05
Ag	ppm	17	29	24	16	43	24	41	40
As	ppm	mdl	0.9	mdl	mdl	mdl	mdl	1.5	mdl
Au	ppm	mdl	mdl	0.20	mdl	0.21	mdl	0.27	0.23
Cd	ppm	3,800	4,280	3,320	3,680	3,090	3,160	3,720	3,200
Cr	ppm	1	4	3	3	4	3	4	2
Cu	ppm	3,400	7,300	8,100	5,200	15,300	8,700	8,700	11,000
Ga	ppm	11.1	64	15	41	11	2.5	6.3	8.4
Hg	ppm	14	11.4	13	12	9.0	10.0	11	11
In	ppm	63	31	74	18	170	255	41	149
Mn	ppm	720	690	1,600	1,660	1,420	1,450	1,050	2,000
Pb	ppm	7	9	13	3.6	70	7	41	50
Sb	ppm	1.3	5.1	1.1	1.0	2.4	mdl	4.0	2.4
Se	ppm	mdl	mdl	9	5	4	4	mdl	mdl
Sn	ppm	190	34	60	65	6	2.3	93	9

Notes: Major elements (Zn, Fe, S) were determined by EMPA and remaining elements were determined by LA-ICP-MS; concentrations below detection limits are reported as mdl

Table 8. Major and Trace Element Compositions of Galena from Vein FF3

Grain Host rock		FF3-GAL1 LS	FF3-GAL2 LS	FF3-GAL3A LS	FF3-GAL3B LS	FF3-GAL4A LS	FF3-GAL4B LS
Pb	wt %	86.36	86.52	86.65	86.57	86.59	86.27
S	wt %	13.43	13.48	13.54	13.57	13.54	13.49
Total		99.79	100.00	100.19	100.14	100.13	99.76
Ag	ppm	1,170	1,220	750	560	970	1,060
As	ppm	20	72	50	80	80	100
Au	ppm	mdl	mdl	mdl	mdl	mdl	0.3
Bi	ppm	1,900	1,980	16	18	1,570	1,560
Cd	ppm	68	58	63	62	44	44
Cr	ppm	mdl	mdl	3	1	6	6
Cu	ppm	mdl	mdl	140	90	mdl	mdl
Fe	ppm	mdl	mdl	700	mdl	mdl	300
In	ppm	0.3	mdl	0.4	0.5	mdl	0.4
Mn	ppm	6	3.5	mdl	4	mdl	mdl
Sb	ppm	370	308	820	590	263	330
Se	ppm	10	5	mdl	mdl	20	mdl
Sn	ppm	3	2	5	7	mdl	3
Te	ppm	13	14	15	20	15	19
Tl	ppm	25.2	25.7	23.2	22.2	25.9	27.0
Zn	ppm	40	mdl	50	50	mdl	30

Notes: Major elements (Pb, S) were determined by EMPA and remaining elements were determined by LA-ICP-MS; concentrations below detection limits are reported as mdl

Unlike the sulfides, tennantite in veins FF2, FF3, and FF5 has variable major element composition in addition to trace elements (Tables 9, 10). This variation is largely the result of substitution of Zn and Fe, which have a strong negative correlation (r^2 = 0.94), although Maydagán et al. (2013) suggested the zonation visible in backscatter electron images may also be the result of Sb-As substitution (Fig. 11). Simply attributing the fluctuating Zn and Fe concentrations to varying physiochemical conditions would not be appropriate as tennantite in veins FF2 and FF5 have low Zn/Fe weight ratios (avg 0.20/1 and 0.01/1, respectively) despite being located at different distances from the center of the deposit (Fig. 2). Vein FF3 was located at a similar distance as FF2, but has tennantite with higher Zn/Fe weight ratios (avg 8.51/1). The main difference

Table 9. Select Major and Trace Element Compositions of Tennantite from Vein Fill

Grain Host rock		FF2-TNT1 MZ	FF2-TNT2 MZ	FF2-TNT4 MZ	FF2-TNT5 MZ	FF3-TNT1A LS	FF3-TNT1B LS	FF3-TNT3 LS	FF5-TNT2 MZ	FF5-TNT3 MZ
Cu	wt %	42.50	43.30	43.20	42.84	42.87	43.26	42.70	45.40	44.65
As	wt %	20.78	21.19	21.48	20.98	20.53	19.94	20.65	20.39	19.78
S	wt %	28.87	28.88	29.04	28.89	28.57	28.61	28.75	28.61	28.46
Zn	wt %	0.82	2.51	mdl	mdl	7.98	7.73	7.19	0.05	mdl
Fe	wt %	6.67	5.30	7.08	7.44	0.72	0.86	1.16	6.29	6.48
Total		99.63	101.18	100.79	100.15	100.67	100.40	100.45	100.74	99.37
Ag	ppm	7,800	3,050	3,560	3,780	330	730	280	350	600
Au	ppm	mdl	mdl	mdl	mdl	mdl	1	mdl	1	3
Bi	ppm	163	950	670	4,400	400	350	84	340	410
Cd	ppm	210	240	11	24	2,200	2,200	2100	mdl	mdl
Cr	ppm	mdl	3	8	10	mdl	11	3	4	6
Ga	ppm	mdl	mdl	140	47	mdl	mdl	mdl	4	12
Hg	ppm	24	22	230	140	33	35	26	21	20
In	ppm	70	66	30	46	52	53	56	mdl	3
Mn	ppm	162	54	1,700	1,460	1,110	1,120	1,180	40	540
Pb	ppm	114	87	3,500	170	50	200	190	60	2,300
Sb	ppm	170	600	3	4	18,500	19,800	7,600	400	1,050
Se	ppm	20	mdl	20	30	20	10	20	90	80
Sn	ppm	166	140	145	76	5	4	30	49	330
Te	ppm	15	mdl	17	16	10	21	10	200	270
Tl	ppm	mdl	mdl	4	1	mdl	mdl	1	129	1,000
V	ppm	mdl	mdl	mdl	mdl	mdl	mdl	mdl	mdl	2

Notes: Major elements (Cu, As, S, Zn, Fe) were determined by EMPA and remaining elements were determined by LA-ICP-MS; concentrations below detection limits are reported as mdl

Table 10. Calculated Formulas for Analyzed Tennantite

Grain	Host rock	Formula
FF2-TNT1	MZ	$(Cu_{9.8}Fe_{1.7}Zn_{0.2}Ag_{0.1})As_{4.0}S_{13.1}$
FF2-TNT2	MZ	$(Cu_{9.9}Fe_{1.4}Zn_{0.6})As_{4.1}S_{13.0}$
FF2-TNT4	MZ	$(Cu_{9.8}Fe_{1.8})As_{4.1}S_{13.1}$
FF2-TNT5	MZ	$(Cu_{9.8}Fe_{1.9}Ag_{0.1})As_{4.1}S_{13.1}$
FF3-TNT1A	LS	$(Cu_{9.8}Zn_{1.8}Fe_{0.2})(As_{4.0}Sb_{0.2})S_{13.0}$
FF3-TNT1B	LS	$(Cu_{9.9}Zn_{1.7}Fe_{0.2})(As_{3.9}Sb_{0.2})S_{13.0}$
FF3-TNT3	LS	$(Cu_{9.8}Zn_{1.6}Fe_{0.3})(As_{4.0}Sb_{0.1})S_{13.1}$
FF5-TNT2	MZ	$(Cu_{10.4}Fe_{1.6})As_{4.0}S_{13.0}$
FF5-TNT3	MZ	$(Cu_{10.3}Fe_{1.7})As_{3.9}S_{13.0}$

Notes: Formulas are calculated based on 29 atoms with data from Table 9

between these veins is host rock: FF2 and FF5 are hosted in monzonite and FF3 is hosted in limestone. This suggests that the host rock not only affects which minerals precipitate but affects their composition as well (Table 10).

Trace elements in tennantite (and textures) suggest that both sphalerite and chalcopyrite were locally replaced by tennantite. The tennantite grains in the limestone-hosted vein (FF3) typically had the highest Cd, Mn, and Sb contents, whereas grains in the monzonite-hosted veins had the highest Sn contents (Table 9). Sphalerite in FF3 has two to three orders of magnitude higher Cd and Mn than chalcopyrite, whereas chalcopyrite has an order of magnitude greater Sn than sphalerite.

Because tennantite is also one of the least abundant minerals and gold concentrations in it are ≤1 to 3 ppm, it cannot be the primary Au host in these rocks which have up to 4.6 ppm Au. In contrast, tennantite can contribute significantly to Ag ore grades with Ag concentrations in tennantite ranging from several 100 to several 1,000 ppm (Table 9). Copper content of tennantite is fairly constant averaging 43.6 ± 1.1 wt %, but its Cu contribution is still overshadowed by the much more abundant chalcopyrite despite its lower Cu concentration (avg 34.1 ± 0.3 wt %).

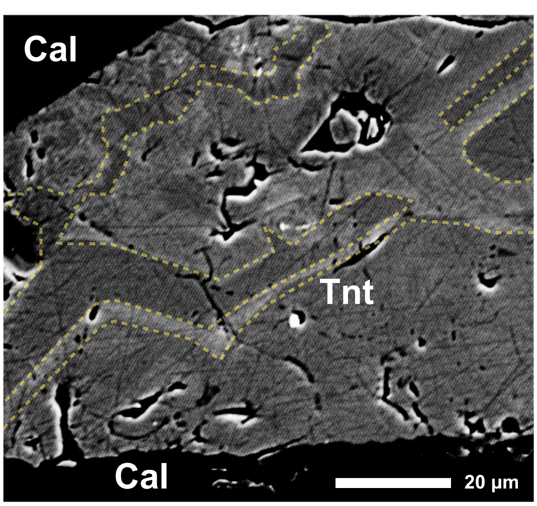

Fig. 11. Backscattered electron image of a tennantite (Tnt) grain from vein FF5 with strong zoning outlined with dashed lines. The more irregular zone closer to the top of the grain may be the result of dissolution, followed by regrowth of additional tennantite with more geometric shape. Cal = calcite.

Sulfur and Oxygen Isotopes

Isotope ratios were measured for quartz (for oxygen isotopes) and pyrite (for sulfur isotopes) from the polymetallic veins. The results are summarized in Table 11.

Sulfur isotopes

The $\delta^{34}S$ values of pyrite in Bingham's polymetallic veins are restricted to a narrow range of 2.3 to 3.4‰. Field and Moore (1971) also studied some of the polymetallic veins and found consistently positive $\delta^{34}S$ values for pyrite (avg 1.6‰, with range 0.4–3.8‰), which decreased with increasing elevation. The latter trend was not observed in our data, although this may be due to the limited range of elevation in our sample set (213 m) relative to theirs (426 m).

The $\delta^{34}S$ values from the pyrites in our study are similar to those in other porphyry-related polymetallic veins in the region, and also those from different parts of the world (Fig. 12). They are also on the high end of $\delta^{34}S_{pyrite}$ values related to porphyry-style mineralization at Bingham (–3.9 to +3.4‰) reported by Field (1966).

Oxygen isotopes

The second and third generations of quartz found by textural analysis (Fig. 6E) and SEM-CL imaging (Fig. 7) were probably incorporated in our bulk quartz separates from each vein, which implies that the $\delta^{18}O$ values reported here are a composite of the two. The first generation of quartz precipitated near vein margins was likely not analyzed, because samples were trimmed where the sulfides ended to avoid wall-rock contamination. The $\delta^{18}O$ values of quartz in the polymetallic veins at Bingham have a mean value of 12.5 ± 0.9‰. This is significantly higher than vein quartz from porphyry-style mineralization at Bingham reported by Bowman et al. (1987), which have a mean value of 9.1 ± 0.6‰ (range 8.3–10.4‰). The $\delta^{18}O$ values from Bowman et al. (1987) become higher with increasing distance from the center of the deposit (r^2 = 0.53). Our sample locations begin roughly where theirs end, and the $\delta^{18}O$ values of quartz from the polymetallic veins appear to be slightly higher in the Fe-Cu zone relative to the Pb-Zn-Cu-Ag-Au zone.

The oxygen isotope composition of quartz is similar to that in other porphyry-related polymetallic vein deposits, except for Butte (Fig. 13). The $\delta^{18}O_{quartz}$ values associated with porphyry-style mineralization at Butte are very similar those at Bingham, yet the trends of $\delta^{18}O_{quartz}$ values from the polymetallic veins associated with each are almost opposite of each other. Garlick and Epstein (1966) noted that a drop in fluid temperature would typically result in an increase of $\delta^{18}O_{quartz}$ values, and attribute the low $\delta^{18}O_{quartz}$ values from Butte's

Table 11. Oxygen and Sulfur Isotopes from Vein Minerals

	Fe-Cu			Pb-Zn-Cu-Ag-Au		
Vein	FF1	FF5	FF7	FF2	FF3	FF4
$\delta^{18}O_{VSMOW}$ quartz	12.0	13.1	14.0	-	11.9	11.5
$\delta^{34}S_{VDCT}$ pyrite	2.9	2.3	2.7	2.8	2.7	3.4

Notes: Vein FF2 did not have sufficient separable quartz in fill material for isotope analysis

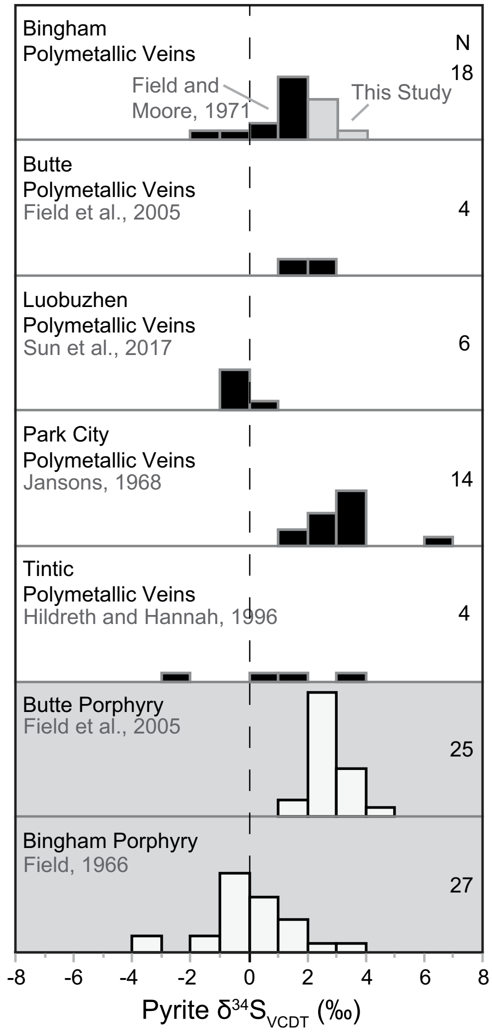

Fig. 12. Histograms of $\delta^{34}S$ values of vein pyrite from various porphyry-related polymetallic veins, as well as data from the porphyry ore zones of Bingham and Butte for additional comparison. The $\delta^{34}S$ values of pyrite from the polymetallic veins are somewhat higher than pyrite from the porphyry-style mineralization (Field, 1966), as well as pyrite from polymetallic veins more distal than our study area (Field and Moore, 1971). Nonetheless, the isotope signature is similar to several porphyry-related polymetallic veins and is indicative of a magmatic sulfur source.

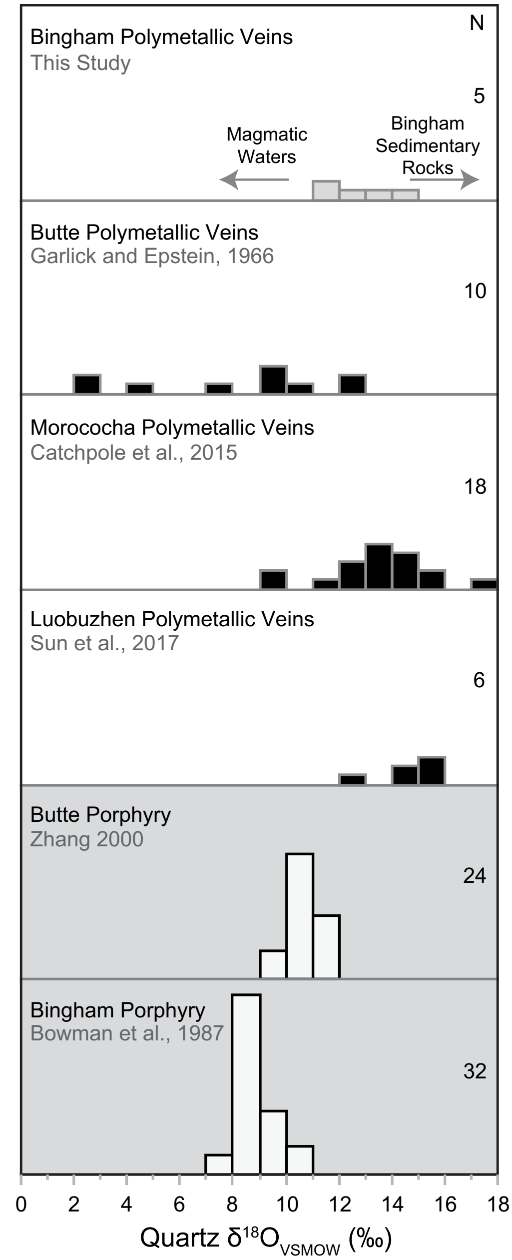

Fig. 13. Histograms of $\delta^{18}O$ values of vein quartz from various porphyry-related polymetallic veins, as well as data from the porphyry ore zones of Bingham and Butte for additional comparison. Quartz grains from the Bingham polymetallic veins have higher $\delta^{18}O$ values and a wider range than quartz associated with porphyry-style mineralization in the deposit (Bowman et al., 1987). The $\delta^{18}O$ values of quartz from the Bingham polymetallic veins show good alignment with quartz from the Morochocha and Luobuzhen polymetallic veins suggesting a common origin, such as the simple cooling of mineralizing fluids. The polymetallic veins at Butte deviate from this trend and have been explained by magmatic fluids mixing with ^{18}O-depleted meteoric water.

polymetallic veins to magmatic fluid mixing with ^{18}O-depleted meteoric water.

Discussion

Source of mineralizing fluids

The $\delta^{34}S_{pyrite}$ values from the polymetallic veins are consistent with derivation from magmatic fluids which typically have values of 0 ± 3‰ (Field et al., 2005), though their slight enrichment could be attributed to the mineralizing fluids cooling. The $\delta^{18}O_{quartz}$ values are not as straightforward, as there are several ways to achieve the elevated values found in polymetallic veins.

Cunningham et al. (2004) explained the elevated $\delta^{18}O$ values in quartzite on the north side of the deposit by calling on the involvement of high ^{18}O water formed in an evaporating crater lake at the summit of the Bingham stratovolcano that drained as groundwater into the mineralizing system. This model presents several challenges, but the data show the need for elevated ^{18}O hydrothermal fluids is not unique to the south side of the deposit where our study was done.

We suggest a different mechanism to explain the high $\delta^{18}O$ values of both quartzite north of pit and quartz in the southern polymetallic veins. The quartzite and carbonate rocks on the north side of the deposit have $\delta^{18}O$ values as high as 18.6 and 29.8‰, respectively (Cunningham et al., 2004). At the time of intrusion, meteoric water would have been heated and convectively flowed outward through these sedimentary rocks facilitating ^{18}O exchange. The result would be enrichment of ^{18}O in the fluids and depletion of ^{18}O in the wall rocks closest to the deposit. In fact, Cunningham et al. (2004) found that $\delta^{18}O$ values in the sedimentary rocks were lowest near the deposit and increased with distance, which aligns with our proposed hypothesis. With increasing distance from the deposit, these high $\delta^{18}O$ exchanged meteoric waters could dominate and mix with pulses of lower $\delta^{18}O$ magmatic fluid. Quartz crystallizing from this mixed hydrothermal fluid would then contain elevated $\delta^{18}O$ values relative to magmatic quartz, or hydrothermal quartz associated with porphyry-style mineralization.

A simpler explanation for the high $\delta^{18}O$ quartz is that isotope fractionation factors are larger at lower temperatures. Evidence for lower temperatures in the vein mineralization includes marcasite found within chalcopyrite veinlets (Fig. 6I), fine-grained quartz veinlets (Fig. 6E), and dull luminescence of that same quartz (Fig. 7). Marcasite typically precipitates at temperatures <240°C (Murowchick and Barnes, 1986), which indicates that the fluids during chalcopyrite precipitation reached this low temperature. The brightness of quartz SEM-CL images has been linked to Ti contents and higher crystallization temperatures (Landtwing et al., 2005; Wark and Watson, 2006; Mercer and Reed, 2013), suggesting low temperature for the dull luminescence of quartz in the Bingham polymetallic veins. Further temperature constraints come from the mineral assemblage of pyrite, chalcopyrite, galena, sphalerite, tennantite (+tetrahedrite), which indicates temperatures <300°C and an intermediate-sulfidation environment (Einaudi et al., 2003; Maydagán et al., 2013; Mercer and Reed, 2013). Fractionation equations of Sharp et al. (2016) yield a 3.5‰ increase in $\delta^{18}O$ of quartz exchanging with the same water when temperature decreases from 350° to 250°C. A 3.5‰ increase to the $\delta^{18}O$ values reported by Bowman et al. (1987) would mostly explain the $\delta^{18}O$ values in the polymetallic veins without a need for a more complicated process. Therefore, we propose cooling fluids as the primary mechanism for $\delta^{18}O$ enrichment, although there may have been some mixing of magmatic fluid with ^{18}O-enriched meteoric water.

Sequence of formation and mineralization

We combine our new results and that of others in order to outline a model of how these polymetallic veins formed at Bingham (Figs. 14, 15). Equigranular monzonite was the first magma to intrude. Overpressure of the magma chamber may have caused the roof of the intrusion to bulge resulting in radial high-angle extensional fracturing (Gruen et al., 2010; Fig. 14B). Steep radial fractures are common close to intrusions, but with increasing distance, they tend to reorient and align with the regional stress field (Boutwell, 1905; Muller and Pollard, 1977; Dilles and Einaudi, 1992; Hildreth and Hannah, 1996; Cammans, 2015). The orientations of the distal polymetallic veins are representations of this phenomenon, having formed as joints at this time with their orientation controlled by far-field stress (Fig. 15).

Intrusion, crystallization, and volatile exsolution of the dike-like, NE-trending quartz monzonite porphyry followed resulting in fluid pressure build-up and the roof to bulge again (Gruen et al., 2010). This dilation could have caused the distal NE-trending joints to dilate and propagate farther out and possibly continued to varying degrees during the emplacement of subsequent intrusions (Fig. 15).

Late mineralizing fluids associated with the emplacement of the narrow quartz latite porphyry dikes were presumably smaller volume than those associated with the significantly larger quartz monzonite porphyry. The extensive joint and fissure network established from the previous intrusions could have acted as conduits for this fluid resulting in long, but narrow, mineralization and alteration (Fig. 14C).

Early quartz-sericite-pyrite alteration, as observed in the Bingham polymetallic veins, is a common feature of distal porphyry-related base metal veins (Einaudi et al., 2003). The silicification of the wall rock could have also acted as a buffer minimizing interaction with the associated wall rocks, allowing the fluids to travel even farther out from their source before precipitating sulfides in the veins (Fig. 14D). Sulfide precipitation appears to have been largely controlled by decreasing temperature due to the lateral zonation of minerals with increasing distance from the center of the associated porphyry deposit. However, host rock also had a significant impact on both the modal abundance of minerals and their compositions. This is most evident in limestone-hosted veins with significantly more Pb-Zn mineralization than the other host rocks (Fig. 5D) and tennantite with high Zn (>7 wt %). These observations lead us to believe the mineralizing fluids were slightly acidic. Limestone would have the greatest reactivity potential to reduce the acidic fluids, thereby shifting sulfide-sulfosalt stability fields and promoting increased mineralization relative to monzonite or quartzite. The Pb-Zn mineralization appears to have followed the quartz-sericite-pyrite alteration, because galena and sphalerite are consistently in the center of the veins (Figs. 5D, 14E). Field and Moore (1971) estimated a temperature of 315°C based on sulfur isotope fractionation of galena-sphalerite pairs from the Galena fissure vein—one of the main polymetallic veins at Bingham (Fig. 2).

Polymetallic veins at Bingham may have formed at the same time as Mo mineralization in the main porphyry orebody. Seo et al. (2012) showed that molybdenite precipitation at Bingham is late following intrusion of the quartz latite porphyry, because Mo veins cut both Cu-Fe sulfide veins and quartz latite porphyry dikes. This is further supported by Re-Os dates on molybdenite (37.0 ± 0.27 Ma) by Chesley et al. (1997) that indicate molybdenite mineralization postdates any of the intrusions at Bingham. Base metal veins are often associated with porphyry deposits rich in Mo (Einaudi et al., 2003). If the Pb-Zn mineralization is associated with Mo mineralization at Bingham, then this would suggest that the fluids responsible for the polymetallic veins occurred after the quartz latite porphyry as well (Fig. 15). A quartz-molybdenite-galena vein recently discovered in the main porphyry Cu-Mo ore zone helps support this idea (Fig. 16).

Chalcopyrite and tennantite precipitation both followed Pb-Zn mineralization after brecciation of early vein fill either by faulting or hydraulic fracturing (Figs. 14F, 15). Both

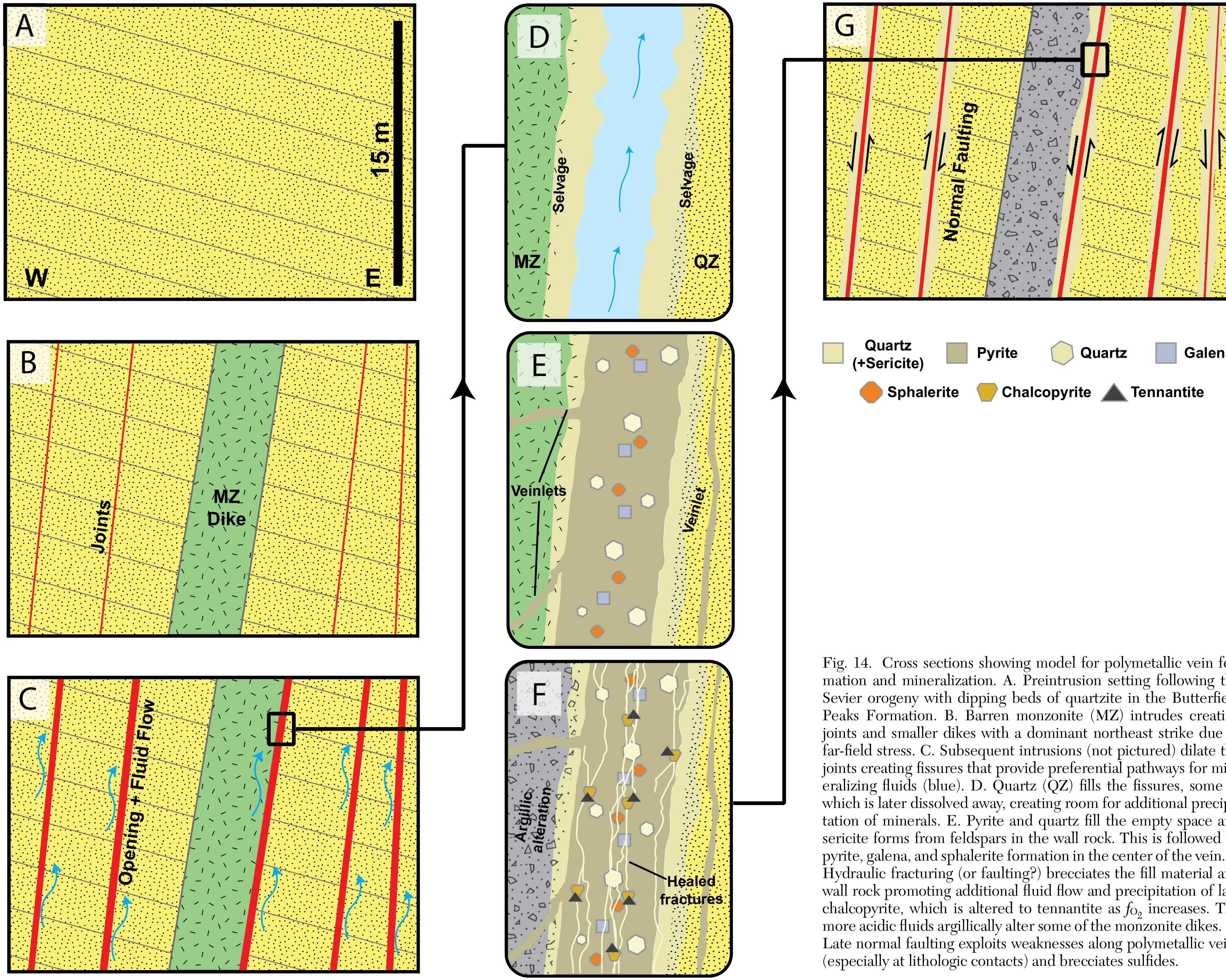

Fig. 14. Cross sections showing model for polymetallic vein formation and mineralization. A. Preintrusion setting following the Sevier orogeny with dipping beds of quartzite in the Butterfield Peaks Formation. B. Barren monzonite (MZ) intrudes creating joints and smaller dikes with a dominant northeast strike due to far-field stress. C. Subsequent intrusions (not pictured) dilate the joints creating fissures that provide preferential pathways for mineralizing fluids (blue). D. Quartz (QZ) fills the fissures, some of which is later dissolved away, creating room for additional precipitation of minerals. E. Pyrite and quartz fill the empty space and sericite forms from feldspars in the wall rock. This is followed by pyrite, galena, and sphalerite formation in the center of the vein. F. Hydraulic fracturing (or faulting?) brecciates the fill material and wall rock promoting additional fluid flow and precipitation of late chalcopyrite, which is altered to tennantite as f_{O_2} increases. The more acidic fluids argillically alter some of the monzonite dikes. G. Late normal faulting exploits weaknesses along polymetallic veins (especially at lithologic contacts) and brecciates sulfides.

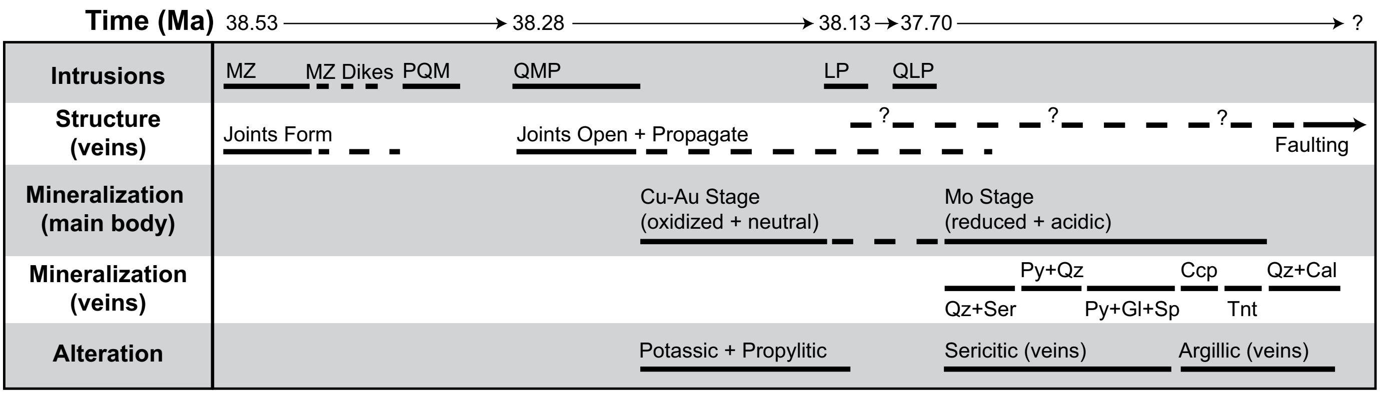

Fig. 15. Timeline of events related to polymetallic vein formation as illustrated in Figure 14 (time component not to scale). Dashed lines indicate episodic activity, or uncertainty if accompanied by a question mark. The formation, opening, and propagation of the northeast joints related to the veins is likely the result of both pressure-driven roof extension during intrusions (Gruen et al., 2010) and hydraulic fracturing by the mineralizing fluids. The relative timing and nature of the main orebody mineralization from Seo et al. (2012), and ages of intrusions from Large (2018).

minerals are present in the third-generation quartz veinlets that heal the fracture networks (Fig. 6J). The more acidic nature of the Mo-stage mineralizing fluids could have remobilized Cu from the main orebody (Seo et al., 2012) and transported it out to the veins. Chalcopyrite filled interstitial space and formed small veinlets sometimes within the fractures of other sulfides. Marcasite in chalcopyrite veinlets in the main veins suggests a pH <5 (Murowchick and Barnes, 1986), which could explain why some of the monzonite dikes that parallel the polymetallic veins are argillically altered (Fig. 14F). Argillic alteration may also be the result of relatively low-temperature fluids inferred for the veins. Tennantite primarily replaces chalcopyrite requiring a sudden increase in f_{O_2}, which could be explained by an influx of meteoric water (Fig. 17). An increase of f_{O_2} would also shift the fluids out of the pyrite stability field, which would free up As for tennantite precipitation that was previously being incorporated into pyrite in the Pb-Zn-Cu-Ag-Au zone (Table 5).

Shearing and brecciation of the veins by normal faulting was the last event (Fig. 14G). Although faulting may have occurred during formation of the veins and intrusions, it definitively occurred after the deposition of the vein fill material, because fault gouge is commonly made up of the sulfides pulverized to clay-sized particles.

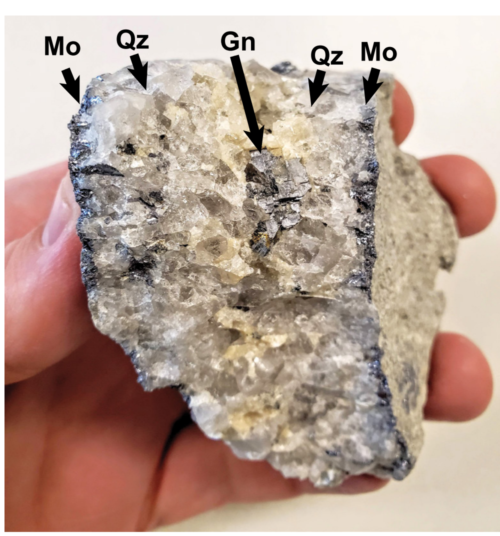

Fig. 16. Hand sample of a quartz-molybdenite-galena vein from the center of the porphyry deposit associated with the polymetallic veins. Galena is rarely found in the center of the deposit, but this recent discovery could suggest Pb-Zn mineralization at Bingham is late following intrusion of the quartz latite porphyry.

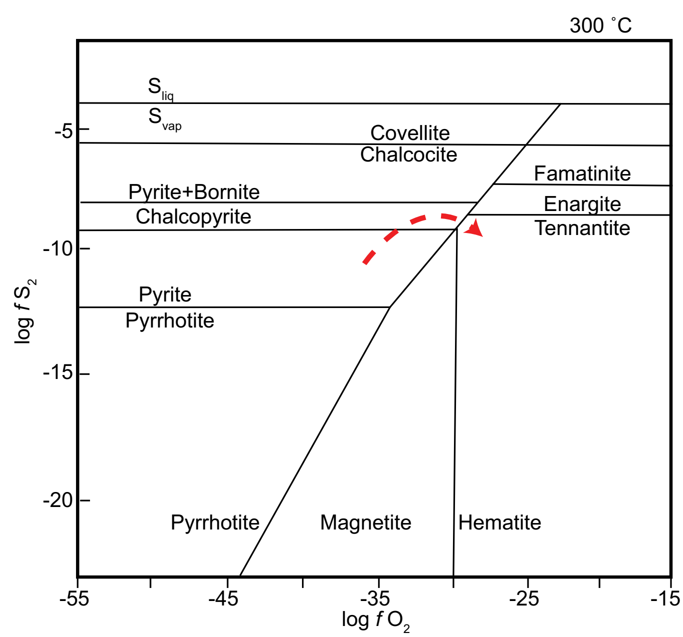

Fig. 17. S_2 vs. O_2 fugacity diagram with stability fields for various Fe-Cu sulfides at 300°C modified from Losada-Calderón et al. (1996). All activities of minerals and H_2O are assumed to equal one. A possible fluid evolution path for the Bingham polymetallic veins is represented by the red dashed line. Pyrite was initially the only sulfide that precipitated, followed by galena and sphalerite. The f_{S_2} increased promoting chalcopyrite precipitation. Textural evidence suggests tennantite formed primarily by alteration of chalcopyrite (Fig. 6I, J). This requires an increase in f_{O_2} in the mineralizing fluids, which may be explained by an influx of meteoric fluids.

Implications for mining and processing polymetallic vein ore

The whole-rock analyses of the sulfide-rich vein fill have maximum values of Au (4.6 ppm), Ag (511 ppm), and Cu (2.3 wt %) that can be ore grade. However, these veins are challenging to economically mine and process. Ore metal grades typically decrease to subeconomic grades within 1.5 m of a vein due to dilution by barren wall rock that cannot be economically separated using bulk mining techniques (Fig. 9). Narrow ore widths explain why the veins were initially mined exclusively underground.

The most advantageous areas for open-pit mining are swarms of closely spaced veins, which are encountered at Bingham (Fig. 4E). All of the major rock types (limestone, monzonite, and quartzite) can have these vein swarms and are capable of hosting mineralized orebodies large enough to economically mine based on recent assays. Complications arise due to the steep dip of the polymetallic veins. The spacing of traditional blasthole drilling at Bingham is approximately 9 m and we have found that mineralization related to the polymetallic veins typically does not exceed 3 m in width. This means not all mineralization associated with these veins can be captured in drill hole assays. The easiest veins to track from bench to bench have shallower dips because this results in more drill holes penetrating the associated mineralization. This occurs near the base of the Jordan and Commercial limestones where polymetallic veins follow bedding rather than cut across it.

The size of the area of vein concentration is not the sole limiting factor. There also needs to be sufficiently high metal concentrations to offset dilution and to compensate for variable metal recovery rates, which are largely related to the composition of the wall rocks. The highest grades of Au, Ag, and Cu are consistently in the limestone-hosted veins followed by monzonite and quartzite veins. This is presumably related to the greater reactivity potential of limestone, which can increase the pH of mineralizing fluids thereby promoting sulfide precipitation. However, metal recovery rates are highest in quartzite due to its simple mineralogy, followed by the monzonite-hosted ore, and finally the limestone-hosted ore. This means that quartzite-hosted veins with relatively low grades can be as economic to mine as higher grade limestone-hosted veins.

Vein grades for Cu, Ag, and Au increase from the center of the deposit outward, but these are also accompanied by higher concentrations of deleterious elements, such as As, Bi, Ni, Pb, Sb, Se, Sn, Te, and Zn (Table 3). These elements persist through downstream processing and their removal requires rigorous efforts to achieve desired purities for the mine's products. For example, Rio Tinto's target purity for silver bars at Bingham is 99.95%. Of the impurities, Bi cannot exceed 10 ppm. However, galena and tennantite, the host minerals of Ag in the veins, can have 1,000s ppm Bi (Tables 8, 9). Even if most of the Bi can be removed, there are still environmental restrictions on the levels of Bi allowed in mine tailings to be considered. These elemental impurities can also build up on the furnace walls of the smelter over time and be transported as fine particles along with SO_2 gas, leading to elevated emissions of heavily monitored elements like Pb. Limestone-hosted veins have the most elevated concentrations of these deleterious elements, which can be orders of magnitude higher than proximal veins hosted in other rock types. This is evident in both the whole-rock and mineralogical compositions (Tables 3, 5). These problematic elements will continue to increase in concentration as distal Pb-Zn mineralization becomes dominant during subsequent expansion of the pit. This will likely require modifications to polymetallic vein ore processing, which could include changes to ore-type blending, processing equipment, and flotation techniques. Additionally, higher cut-off grades may need to be considered to account for the increased efforts required to remove the abundant deleterious elements.

Conclusions

The Pb-Zn-Cu-Ag-Au veins at Bingham show a systemic metal zonation outward from the center of the main porphyry deposit from Fe-Cu to Pb-Zn-Cu-Ag-Au. This shift in metal abundances is associated with higher concentrations of chalcophile trace elements in pyrite, chalcopyrite, galena, sphalerite, and tennantite. Associated alteration is predominantly sericitic, which is superimposed on and is physically and chemically distinct from the widespread propylitic alteration in the area. Mineralization typically does not extend >1.5 m into wall rock, although. more intensely brecciated quartzite wall rock can form wider ore zones.

The mineralogical host of Au could not be determined, although pyrite appears to be the dominant contributor simply due to its overwhelming modal abundance. Galena and tennantite host most Ag, and Cu is hosted primarily in chalcopyrite, tennantite, and sphalerite. The limestone-hosted veins have substantially larger amounts of galena and sphalerite (and higher grades of Cu, Au, and Ag when considering a larger vein sample set) relative to quartzite- or monzonite-hosted veins. This is presumably due to the greater buffering capacity of limestone with acidic mineralizing fluids.

Stable isotope data suggest that the polymetallic veins formed from magmatic fluids possibly mixed with meteoric water. This is shown by the magmatic sulfur isotope composition of pyrite ($\delta^{34}S$ = 2.3–3.4‰), and heavy oxygen isotopic composition of quartz ($\delta^{18}O$ = 11.5–14.0‰). These isotope signatures are not unique to the Bingham porphyry system and are likely the result of cooling of pulses of magmatic fluids. Mixing of the magmatic water with meteoric water would further oxidize and cool the fluids while diluting the ligands and promoting sulfide deposition. The meteoric water could have been enriched in ^{18}O by heating and convection through the ^{18}O-rich sedimentary rocks following intrusion of magma, but this is not required to explain the high $\delta^{18}O$ values of quartz in the polymetallic veins.

Polymetallic veins are commonly associated with porphyry copper deposits; thus, the results of this study can be used to better understand similar deposits and improve efficiency in mining, processing, and exploration. Expansion of open-pit porphyry copper mines is accompanied by tremendous expense, which can be partly alleviated by exploiting more distal vein-controlled deposits that can form sizable and economic ore lodes.

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